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Pattern formation in a predator–prey model on network and non-network environments[☆]

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ABSTRACT

Pattern formation in mathematical modeling of biology has been a popular topic for a long time. In this study, we focus on possible patterns driven by Turing instability under the prey–predator model of nonlinear reaction–diffusion type in 2D space with the Allee effect and intra-species competition. We investigate pattern formation in predator–prey systems on both continuous media and network frameworks, highlighting how diffusion in continuous space and its Laplacian analogue on networks shape the emergence of Turing patterns. We derive the model from scratch, discuss three separate cases, and analyze the dynamics of both temporal and spatiotemporal dynamics of the models. Under circumstances where the Turing pattern does not exist, we present detailed proofs and use numerical simulation to demonstrate temporal Hopf bifurcation and patchy invasion. On the other hand, if the analytical conditions for the Turing pattern can be satisfied, we display patterns in our numerical simulations and give corresponding biological interpretations at the end.

1. Introduction

Over the past two decades, studies have provided important information on spatial ecological systems formed by predator–prey interactions. Therefore, research on the distribution of interacting species within a changing ecological environment can be valuable and informative in terms of mathematical and biological aspects. Previous studies have proposed various interesting findings. For instance, Haque [1] has demonstrated that the intra-species competition among predators benefits many predator–prey systems in terms of keeping the system's positive equilibrium points stable. In addition, the 'paradox of enrichment' phenomenon obtained by Roy and Chattopadhyay [2] suggests that increasing the environmental carrying capacity for the prey population will lead to an increase in the predator density in equilibrium points but not in prey density, thereby will destabilize the system eventually.

In particular, in the history of the development of predator–prey models, 'trophic function' or 'functional response' has been thought of as a key factor in determining the stability and bifurcation dynamics of the system [1]. While a variety of forms of the term 'trophic function' have been suggested, this paper use the definition that is the function of prey density, which was labeled by as 'prey-dependence' [3]. In addition, pattern formations, which have been discussed in [4], are of great importance in mathematical biology, especially in the mathematical modeling of biological interactions in nature.

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This paper aims to study the spatiotemporal behaviors of the reaction–diffusion predator–prey model with the Allee effect under a uniform environment where the system parameters do not depend on space or time. A combination of quantitative and qualitative approaches is used in the model analysis. Recently, there has been growing interest in extending classical reaction–diffusion models from continuous Euclidean domains to complex networks and graphs. Unlike regular spatial habitats, real ecological and biological systems often display discrete, heterogeneous interaction structures for instance, fragmented landscapes, food webs, or neural networks. On such networks, diffusion is represented through fluxes along edges, with the Laplacian operator replaced by its graph analogue. This framework has revealed that Turing-like instabilities and spatiotemporal patterns can emerge even in non-geometric settings, with the network topology strongly shaping the onset and type of patterns that form [5–7]. Incorporating network-based diffusion into predator–prey systems therefore provides a powerful way to capture realistic dispersal pathways and heterogeneities that cannot be explained by classical continuous-space models [8,9].

In this paper, we examine the prey–predator model, providing a biological interpretation of each term and constructing the model incrementally to reach the final formulation.

Initially, we consider the temporal variation in prey and predator densities. It is essential to model the prey growth term and the interaction, specifically the hunting process, between prey and predators. We introduce r as the prey growth rate, m as the consumption rate, d as the predator death rate, and h as the intra-species competition rate among predators [1,10]. We posit that changes in prey density may result from intrinsic growth and the hunting process. Conversely, changes in predator density may arise from the hunting process, predator mortality, and intra-species competition.

Therefore, our model in the first stage is a system of ODEs :

$$\frac{dx}{dt} = rx - mxy, \tag{1a}$$
$$\frac{dy}{dt} = mxy - dy - hy^2. \tag{1b}$$

However, this formulation raises certain issues regarding biological realism. In particular, as x becomes large, the intrinsic growth term rx increases without bound, and the interaction term mxy likewise grows indefinitely. To obtain a more realistic description, we instead adopt a logistic growth model for the prey population, where k represents the environmental carrying capacity for the prey. In addition, we introduce a and c as prey saturation parameters, which impose saturation effects on the prey–predator interaction and thereby constrain the intensity of predation at high prey densities [11,12]. In this way, our model becomes:

$$\frac{dx}{dt} = rx(1-\frac{x}{k}) - \frac{mxy}{ax+c}, \tag{2a}$$
$$\frac{dy}{dt} = \frac{emxy}{ax+c} - dy - hy^2. \tag{2b}$$

As x approaches k, the growth term converges to 0, which prevents the solution from blowing up. Moreover, for large values of x, the interaction term $\frac{mxy}{ax+c}$ remains bounded. Consequently, the model is now more biologically realistic. However, this formulation remains incomplete from a biological perspective. We now introduce the Allee effect in our model [13], which will modify the growth term to $rx(1-\frac{x}{k})(x-n_1)$, where n_1 is the Allee constant. Then the model becomes:

$$\frac{dx}{dt} = rx(1-\frac{x}{k})(x-n_1) - \frac{mxy}{ax+c}, \tag{3a}$$
$$\frac{dy}{dt} = \frac{emxy}{ax+c} - dy - hy^2. \tag{3b}$$

The details of the Allee effect are discussed in the following sections.

In this work, our primary objective is to systematically identify and characterize the range of patterns that can arise within the framework of our proposed model. Therefore, we consider not only the temporal change of prey and predator density but also the spatial distribution of the prey and predator. We include diffusion terms in 2 dimensions in our model [14], where D_x and D_y are the diffusion coefficients of the prey and predator, respectively. We also assume that no species enters or leaves the well-defined biological environment. Finally, we have arrived at our general model:

$$\frac{\partial x}{\partial t} = rx(1-\frac{x}{k})(x-n_1) - \frac{mxy}{ax+c} + D_x\nabla^2 x, \tag{4a}$$
$$\frac{\partial y}{\partial t} = \frac{emxy}{ax+c} - dy - hy^2 + D_y\nabla^2 y. \tag{4b}$$

This system adheres to positive initial conditions

$$x(s,0) = v_1 \geq 0, \quad y(s,0) = v_2 \geq 0, \tag{2.5}$$

where $s=(u,v)\in\Omega\subset\mathbb{R}^2$ represents the two-dimensional space, $t\geq 0$, and it is subjected to zero-flux Neumann boundary conditions given by

$$\frac{\partial x}{\partial \eta} = \frac{\partial y}{\partial \eta} = 0, \tag{5}$$

where η symbolizes the unit outward normal vector on the boundary $\partial\Omega$. The summary of variables and parameters used in the predator–prey reaction–diffusion model are mentioned in the Table 1.

Table 1
Summary of variables and parameters used in the predator–prey reaction–diffusion model.

Symbol	Description
$x(t)$	Prey population density at time t
$y(t)$	Predator population density at time t
r	Intrinsic growth rate of the prey
k	Environmental carrying capacity of the prey
m	Predation (consumption) rate
e	Conversion efficiency of prey into predator biomass
d	Natural death rate of the predator
n_1	Allee threshold constant
a	Prey saturation constant in the functional response
h	Intra-species competition rate among predators
c	Saturation constant in the functional response
D_x	Diffusion coefficient of the prey
D_y	Diffusion coefficient of the predator
∇^2	Laplacian operator in two-dimensional space

Although diffusion-driven instabilities on networks have been widely studied within the framework of Master Stability Functions (MSF) and related spectral approaches [15], the primary goal of the present work is different. Rather than developing a general instability framework, we focus on how specific ecological mechanisms namely the Allee effect, prey saturation, and predator intra-species competition-shape spatial dynamics in predator–prey systems under both continuous and network-based diffusion. In particular, our objective is to examine how these biologically motivated reaction terms influence the emergence of Turing patterns and patchy invasion structures, and how these phenomena are modified when dispersal occurs on discrete network topologies. From this perspective, the network formulation is used as a complementary framework to investigate the influence of connectivity structure on spatial ecological dynamics, rather than as a platform for developing a general spectral theory of network instabilities.

The network analogue of the predator–prey system can therefore be written as

$$\frac{dx_i}{dt} = rx_i\left(1 - \frac{x_i}{k}\right) - \frac{mx_iy_i}{ax_i + c} + D_x\sum_{j=1}^{N} L_{ij}x_j, \tag{6a}$$

$$\frac{dy_i}{dt} = \frac{emx_iy_i}{ax_i + c} - dy_i - hy_i^2 + D_y\sum_{j=1}^{N} L_{ij}y_j, \quad i = 1, \dots, N, \tag{6b}$$

where x_i and y_i represent the populations at node i, D_x and D_y are the diffusion coefficients, and L_{ij} is the Laplacian matrix of the network.

The remainder of the paper is organized as follows: In Section 2, we non-dimensionalize the general model for the convenience of numerical simulations in the following sections. Section 3 deals with analyzing the Allee effect term in our general model. Section 4 outlines the principle assumptions of the prey–predator model. The following three sections present the findings of analyzing the model's dynamics and possible patterns driven by Turing instability through three separate cases, including $h = 0$ and $d = 0$, and none of them are equal to 0. In Section 5, we perform stability analysis and prove the existence of Hopf bifurcation [16]. In addition, we prove the non-existence of the Turing pattern in this model by stating that the analytical conditions for Turing instability cannot be satisfied in this model. Then, following the investigation of patchy invasion by [17,18], we demonstrate patchy invasion in the same model and give biological interpretation to our parameters through graphic explanation. In Section 6, we pay attention to the satisfaction of analytical conditions of patchy invasion and possible patterns driven by Turing instability. This serves as an introductory part for Section 7, which analyses the general model we derive from the mathematical model part. In this section, we use numerical simulation to demonstrate various patterns and give tables of numerical simulation parameters and similar pattern parameters. In Section 8, we extend the study of Turing patterns to network-structured habitats. Using the Laplacian matrix as the discrete analogue of diffusion, we show how network topology influences the onset and form of spatial patterns. Section 9 concludes the paper with biological interpretations for the previous numerical simulation results, including patchy invasion and Turing pattern.

2. Non-dimensionalization of the model

Before diving into this complex model, we need to first non-dimensionalize the model to simplify the calculation process. By setting the following new variables,

$$X = \frac{x}{x_0}, Y = \frac{y}{y_0}, \tau = \frac{t}{t_0}, X_{axis} = \frac{x_{axis}}{x_1}, Y_{axis} = \frac{y_{axis}}{y_1}$$

we get the non-dimensionalized model:

$$\frac{\partial X}{\partial \tau} = \frac{cr}{em}X(1 - X)(X - \frac{n_1}{k}) - \frac{XY}{\frac{ak}{c}X + 1} + (\frac{\partial^2 X}{\partial^2 X_{axis}} + \frac{\partial^2 X}{\partial^2 Y_{axis}}),$$

$$\frac{\partial Y}{\partial \tau} = \frac{XY}{\frac{ak}{c}X + 1} - \frac{cd}{emk}Y - \frac{ch}{emk}Y^2 + \frac{D_y}{D_x}\left(\frac{\partial^2 Y}{\partial^2 X_{axis}} + \frac{\partial^2 Y}{\partial^2 Y_{axis}}\right)$$

where $t_0 = \frac{c}{emk}, x_0 = k, y_0 = e, x_1 = y_1\sqrt{\frac{cD_x}{emk}}$

Let $A = \frac{cr}{em}$, $B = \frac{n_1}{k}$, $S = \frac{ak}{c}$, $D = \frac{D_y}{D_x}$, $E = \frac{cd}{emk}$ and $F = \frac{ch}{emk}$. In addition, by replacing τ with t, and X, Y with x, y, we arrive at the following system of PDEs:

$$\frac{\partial x}{\partial t} = Ax(1-x)(x-B) - \frac{xy}{Sx+1} + \nabla^2 x, \tag{7a}$$

$$\frac{\partial y}{\partial t} = \frac{xy}{Sx+1} - Ey - Fy^2 + D\nabla^2 y, \tag{7b}$$

The network analogue for this model is

$$\frac{dx_i}{dt} = A\,x_i\,(1-x_i)\,(x_i - B) - \frac{x_i y_i}{Sx_i + 1} + \sum_{j=1}^{N} L_{ij} x_j, \tag{8a}$$

$$\frac{dy_i}{dt} = \frac{x_i y_i}{Sx_i + 1} - E\,y_i - F\,y_i^2 + D\sum_{j=1}^{N} L_{ij} y_j, \quad i = 1, \dots, N, \tag{8b}$$

where x_i and y_i denote the state variables at node i in the network, and L_{ij} is the Laplacian matrix of the network. The system extends the classical reaction–diffusion model to discrete networked domains, allowing investigation of pattern formation and dynamics on complex connectivity structures.

3. Analysis of the Allee effect in the prey dynamics

In this section, we examine the role of the Allee effect in the prey dynamics. The Allee effect refers to the phenomenon in which the survival and reproduction success of a population decreases at low population densities. In ecological systems, two forms of the Allee effect are commonly distinguished [19]. A strong Allee effect occurs when the population growth rate becomes negative below a critical population threshold, which may ultimately lead to extinction. In contrast, a weak Allee effect describes a situation in which the growth rate remains positive at low densities but is reduced relative to its maximum value.

To analyze the Allee effect in isolation, we consider the prey dynamics in the absence of predation and spatial diffusion. Setting $m = 0$ and neglecting diffusion $D_x = 0$, Eq. (4a) reduces to

$$\frac{dx}{dt} = rx\left(1 - \frac{x}{k}\right)(x - n_1). \tag{9}$$

Here r denotes the intrinsic growth rate of the prey population, k represents the environmental carrying capacity, and n_1 denotes the Allee threshold. The condition $D_x = 0$ implies that only temporal population dynamics are considered.

The equilibrium points are obtained by setting $\frac{dx}{dt} = 0$, yielding $x^* = \{0, n_1, k\}$. The stability of these equilibria can be analyzed by computing

$$\frac{d}{dx}\left(rx\left(1 - \frac{x}{k}\right)(x - n_1)\right) = rx\left(1 - \frac{x}{k}\right) + r\left(1 - \frac{x}{k}\right)(x - n_1) - \frac{rx(x - n_1)}{k}. \tag{10}$$

Since n_1 represents the Allee threshold and k denotes the environmental carrying capacity, we assume $n_1 < k$. Evaluating the derivative at the equilibrium points $x^* = \{0, n_1, k\}$ gives

$$\left.\frac{d}{dx}\right|_{x=0} = -rn_1 < 0, \qquad \left.\frac{d}{dx}\right|_{x=n_1} = \frac{n_1 r(k - n_1)}{k} > 0, \qquad \left.\frac{d}{dx}\right|_{x=k} = r(n_1 - k) < 0.$$

Hence $x^* = 0$ and $x^* = k$ are locally stable equilibria, while $x^* = n_1$ is unstable. The point $x = n_1$ therefore represents the Allee threshold separating extinction dynamics from population persistence.

Fig. 1 illustrates the influence of the Allee effect on prey population growth. At very low population densities the growth rate becomes negative. As the population increases, the growth rate initially rises and eventually vanishes at the carrying capacity $x = k$. The Allee effect can significantly influence the stability of the system and may induce nonlinear behaviors such as periodic oscillations or extinction. To illustrate these dynamics, we consider model Eq. (3a)–(3b). consider model Eq. (3a)–(3b). Fig. 2 shows that the model may exhibit three distinct dynamical regimes. In the first case, both populations converge to a stable equilibrium. In the second, prey and predator densities exhibit sustained oscillations. In the third case, both species go extinct, with the prey population collapsing earlier than the predator population. Among these outcomes, we particularly focus on periodic oscillations, as they align with the natural life cycles of biological populations. When the prey population exceeds the Allee threshold, it tends to grow toward the carrying capacity of the environment. However, the additional term $\frac{mxy}{ax+c}$ introduces predation pressure, reducing prey abundance and thereby complicating the Allee effect. We hypothesize that oscillatory behavior emerges from this feedback mechanism: the growth of the prey population supports predator growth, which in turn intensifies predation pressure and reduces the prey population. As the prey declines, predation pressure weakens, allowing the population to recover. This dynamic interplay between the Allee effect and predation pressure results in oscillatory population cycles. The model is analyzed under several biologically motivated assumptions. All parameters $(r, m, k, n_1, a, e, d, h, D_x, D_y)$ are assumed to be positive, ensuring biologically

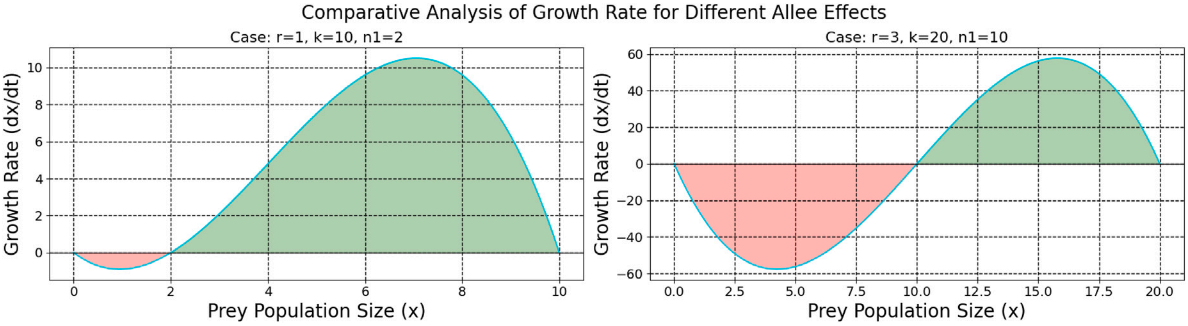


Fig. 1. Illustration of the Allee effect. There exists a threshold $x_s > 0$ such that when $x > x_s$ the growth rate satisfies $\frac{dx}{dt} \geq 0$, whereas for $x < x_s$ the growth rate $\frac{dx}{dt} < 0$.

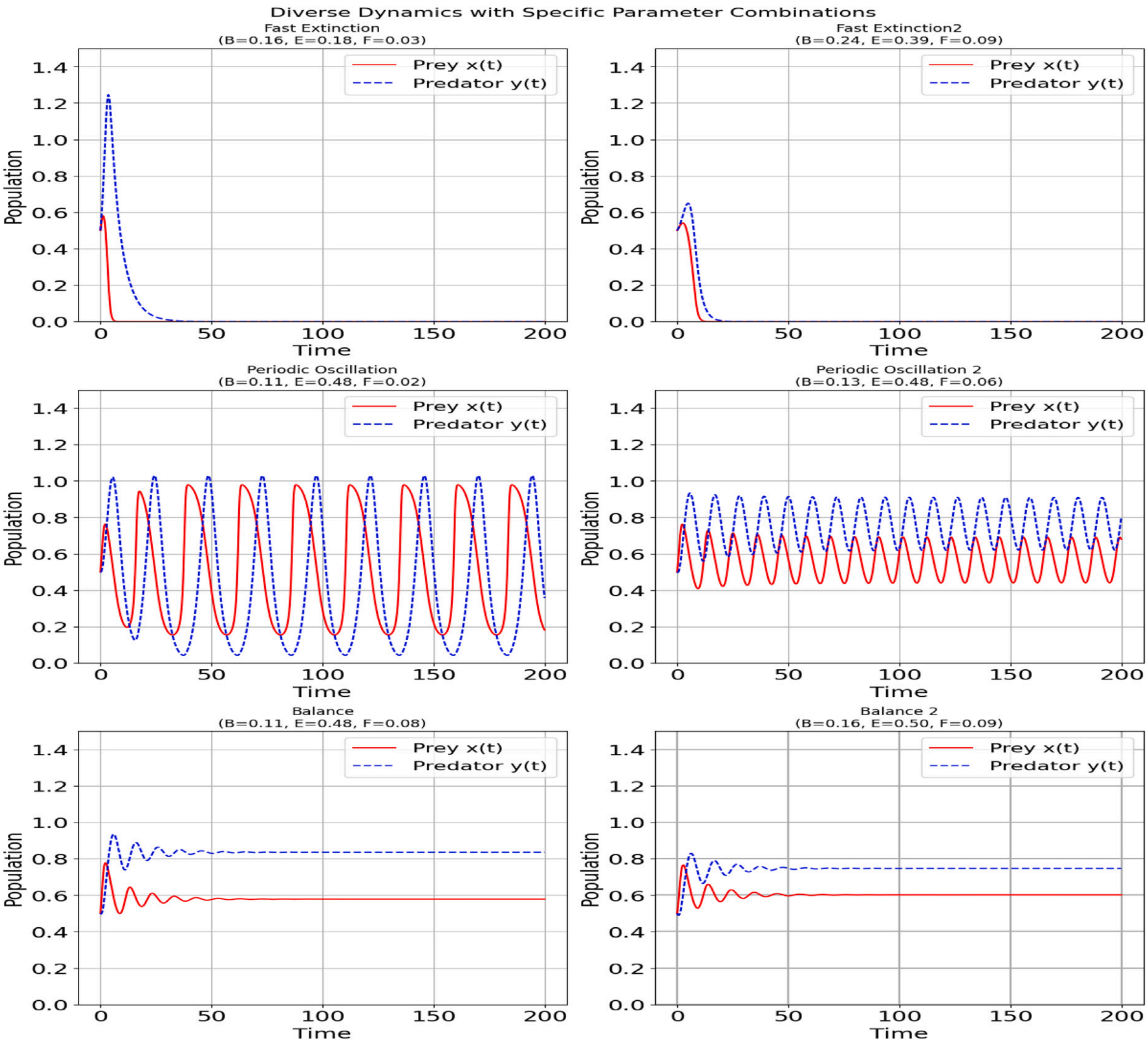


Fig. 2. Representative dynamical behaviors of the system under different parameter values: stable equilibrium, periodic oscillations, and extinction.

meaningful interpretations of the system. Consequently, the non-dimensional parameters A, B, S, D, E, and F are also positive. In addition, since n_1 represents the Allee threshold below which the prey population cannot sustain positive growth, biological realism requires that the Allee threshold be smaller than the environmental carrying capacity of the prey. Hence we assume $n_1 < k$, which implies

$$B = \frac{n_1}{k} < 1.$$

4. Case 1: Model without predator intra-species competition ($h = 0$)

We first consider the case $h = 0$ in our original model, which implies $F = \frac{ch}{emk} = 0$. Biologically, this corresponds to the absence of intra-species competition among predators. The model without diffusion reduces to the following system of ODEs:

$$\frac{dx}{dt} = Ax(1-x)(x-B) - \frac{xy}{Sx+1}, \tag{11a}$$

$$\frac{dy}{dt} = \frac{xy}{Sx+1} - Ey. \tag{11b}$$

To examine the dynamics and the potential for pattern formation, we first study the stability of the equilibrium points. The Jacobian matrix of the system is given by

$$J = \begin{pmatrix} J_{11} & -\frac{x}{Sx+1} \\ -\frac{Sxy}{(Sx+1)^2} + \frac{y}{Sx+1} & -E + \frac{x}{Sx+1} \end{pmatrix}.$$

where $J_{11} = Ax(1-x) - Ax(-B+x) + A(1-x)(-B+x) + \frac{Sxy}{(Sx+1)^2} - \frac{y}{Sx+1}$. The equilibrium points of the system are $E_1 = (0,0)$, $E_2 = (1,0)$, $E_3 = (B,0)$, and

$$E_4 = \left(-\frac{E}{ES-1}, \frac{A(ES+E-1)(BES-B+E)}{(ES-1)^3}\right).$$

We are primarily interested in E_4, since it is the only equilibrium that can yield positive x and y values, which are necessary for pattern formation. However, the other equilibria also provide useful information. At $E_1 = (0,0)$, the Jacobian simplifies to

$$J(E_1) = \begin{pmatrix} -AB & 0 \\ 0 & -E \end{pmatrix},$$

which gives

$$\begin{cases} \operatorname{tr}(J_{E_1}) = -AB - E < 0, \\ \det(J_{E_1}) = ABE > 0. \end{cases}$$

Thus, both eigenvalues are negative, and $(0,0)$ is always locally stable. From this, we conclude that E_4 cannot be globally stable. At E_4, the Jacobian matrix of the model evaluated at this fixed point is

$$J_{E_4} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix},$$

where a_{11}, a_{12}, a_{21}, a_{22} are provided in Appendix A.

The conditions for E_4 to be locally stable are given by

$$\begin{cases} a_{11} + a_{22} < 0, \\ a_{11}a_{22} - a_{21}a_{12} > 0. \end{cases} \tag{12}$$

If these conditions are satisfied, the equilibrium point E_4 is locally stable.

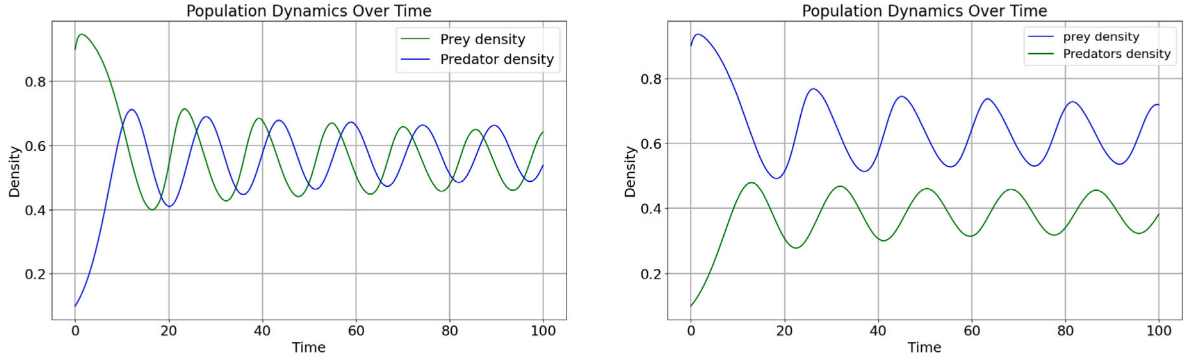
4.1. Hopf bifurcation analysis

An important question in prey–predator dynamics is whether the system admits sustained oscillatory solutions. Such behavior may arise through a Hopf bifurcation, which occurs when a pair of complex conjugate eigenvalues of the Jacobian matrix crosses the imaginary axis. The characteristic equation of the Jacobian matrix evaluated at the interior equilibrium E_4 is

$$\lambda^2 - \operatorname{tr}(J_{E_4})\lambda + \det(J_{E_4}) = 0.$$

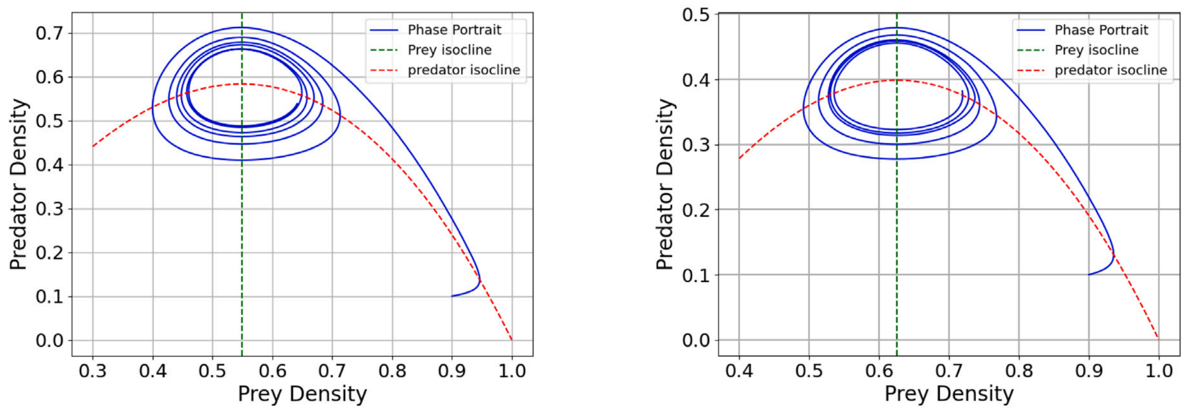
We consider $B = \frac{n_1}{k}$ as the bifurcation parameter. A Hopf bifurcation occurs when the eigenvalues become purely imaginary, which requires $\operatorname{tr}(J_{E_4}) = 0$ while $\det(J_{E_4}) > 0$.

For the parameter values $A = 2$, $S = 0.4$, and $E = 0.45$, the above condition yields the critical bifurcation value $B = 0.0191801213786787$. Moreover, the real parts of the eigenvalues change sign as B passes through this threshold, indicating the occurrence of a Hopf bifurcation. Consequently, periodic oscillations emerge in the neighborhood of the interior equilibrium when B is chosen near this critical value. The resulting periodic solutions are illustrated in Fig. 3 and their phase portrait in Fig. 4.



(a) $A = 2,\ B = 0.019,\ S = 0.4,\ E = 0.45$ (b) $A = 2,\ B = 0.2,\ S = 0.4,\ E = 0.5$

Fig. 3. Time series illustrating oscillatory dynamics arising from a Hopf bifurcation with initial conditions $x_0 = 0.9$ and $y_0 = 0.1$.



(a) Phase portrait corresponding to Fig. 3(a) (b) Phase portrait corresponding to Fig. 3(b)

Fig. 4. Phase plane plots for prey and predator densities. The blue curve represents periodic solutions. The red dotted line indicates the predator isocline, and the green dotted line indicates the prey isocline. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.2. Diffusion-driven pattern formation

The necessary and sufficient conditions for Turing instability are given by the following system

$$\begin{cases} ES < 1, \dfrac{A(ES + E - 1)(BES - B + E)}{(ES - 1)^3} > 0 \\ a_{11} + a_{22} < 0, \\ a_{11}a_{22} - a_{21}a_{12} > 0, \\ D\,a_{11} + a_{22} > 2\sqrt{D}\sqrt{\det(J_{E_4})}. \end{cases} \tag{13}$$

We are able to establish the following result.

Theorem 1. *The model (11) does not exhibit Turing instability, as the fourth condition in (13) cannot be satisfied at any positive interior equilibrium.*

Proof. See in Appendix B. □

4.3. Patchy invasion dynamics

Patchy invasion refers to a spatial invasion process in which the spreading population does not form a smooth and continuous invasion front but instead breaks into irregular and disconnected patches. In such cases, the invading species expands through

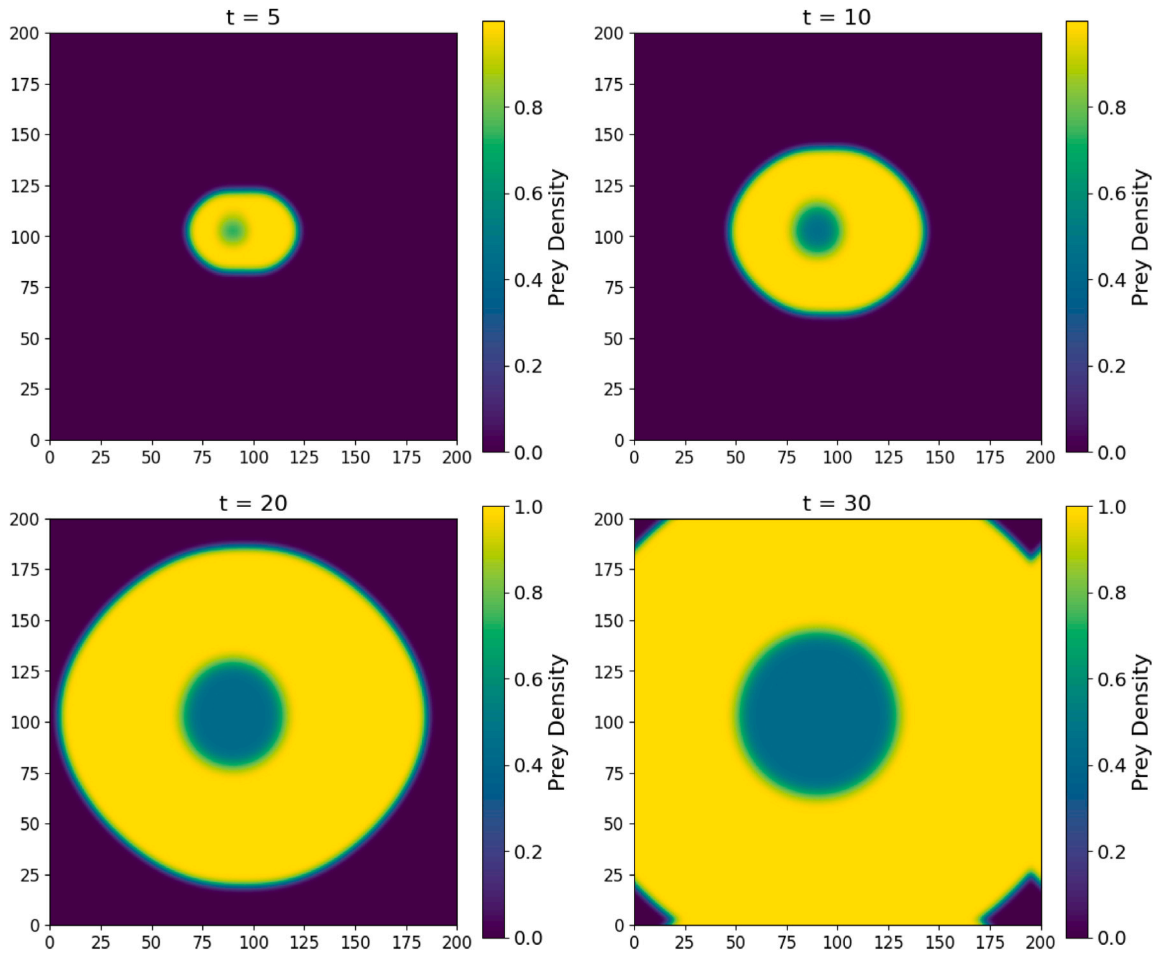


Fig. 5. Spatial distribution of prey density in the absence of the Allee effect. The horizontal and vertical axes correspond to the spatial coordinates x and y. Initial conditions: $x_0 = 1.2$, $y_0 = 0.2$, $x_{11} = 85$, $x_{12} = 105$, $y_{11} = 100$, $y_{12} = 105$, $x_{21} = 85$, $x_{22} = 95$, $y_{21} = 95$, $y_{22} = 115$. Parameter values: $A = 4$, $B = 0.22$, $S = 0.15$, $E = 0.4$, $D = 1.2$.

the formation of localized population clusters that appear and disappear over space and time. This phenomenon differs from classical Turing patterns, which arise from diffusion-driven instability around a spatially homogeneous equilibrium. In contrast, patchy invasion is typically associated with transient spatial dynamics during the invasion process and is strongly influenced by nonlinear ecological mechanisms such as Allee effects or predator–prey interactions [17,18]. In this subsection, we reproduce the patchy invasion phenomenon using the same initial conditions but with parameter values modified from those in [17]. We then further vary the parameters and provide our own biological interpretation of them by comparing the resulting numerical simulations. Specifically, we take the initial condition to be $x = x_0, y = y_0$, and we assume that $x = 0$ whenever x does not satisfy $x_{11} < x < x_{12}$ and $y_{11} < y < y_{12}$, and $y = 0$ whenever y does not satisfy $x_{21} < x < x_{22}$ and $y_{21} < y < y_{22}$.

Fig. 5 presents the spatial pattern in the absence of an Allee effect. As illustrated, once the invasion begins, the prey forms an expanding circular front, while the population density near the center is relatively low due to intense predation. The prey are pursued by the predator and spread outward from the center in an approximately uniform manner. This behavior is consistent with what is typically expected for an invasion process without an Allee effect.

Fig. 6 presents the invasion dynamics in the presence of an Allee effect. As illustrated, in contrast to the standard invasion scenario without an Allee effect, the incorporation of the Allee effect leads to the emergence of spatially discrete patches. Initially, the prey still forms an approximately circular invasion front, but this front subsequently fragments into multiple clusters. Thereafter, the spread of the prey becomes spatially heterogeneous, producing disconnected patches as the invasion progresses.

A direct comparison between the two scenarios—using identical parameter values with and without the Allee effect demonstrates that the Allee effect transforms the invasion pattern from a smooth, circular front into an irregular, patchy structure. To further elucidate these differences, one can also examine phase plane diagrams plotting predator density against prey density.

Next, we carry out further numerical simulations to obtain additional patchy invasion patterns. The occurrence of patchy invasion is highly sensitive to changes in parameter values. By slightly modifying these parameters, we identify several other intriguing scenarios that merit presentation and discussion. These cases will be shown in Fig. 7.

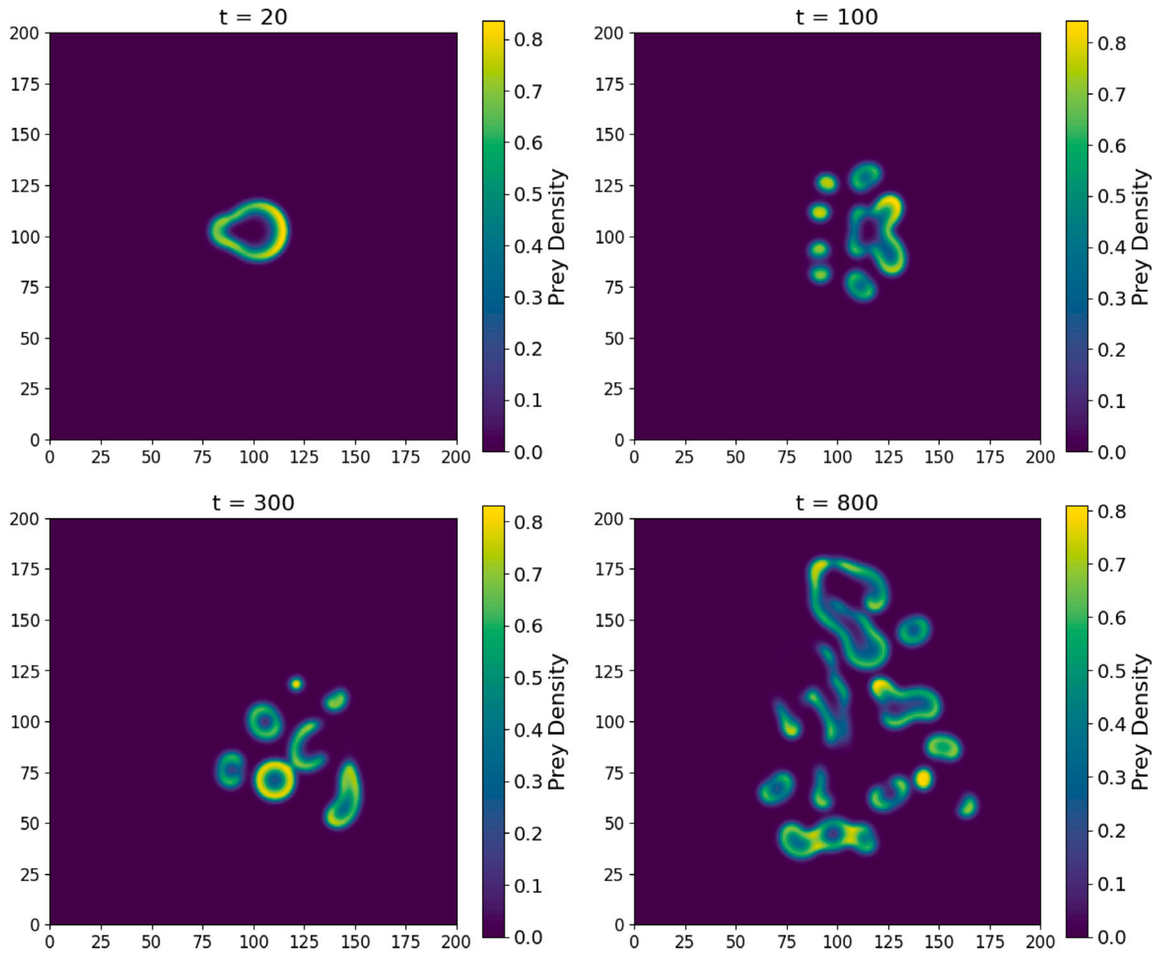


Fig. 6. Spatial distribution of prey density in the presence of the Allee effect. The horizontal and vertical axes represent the spatial coordinates x and y. Parameter values: $A = 4$, $B = 0.22$, $S = 0.15$, $E = 0.4$, $D = 1.2$.

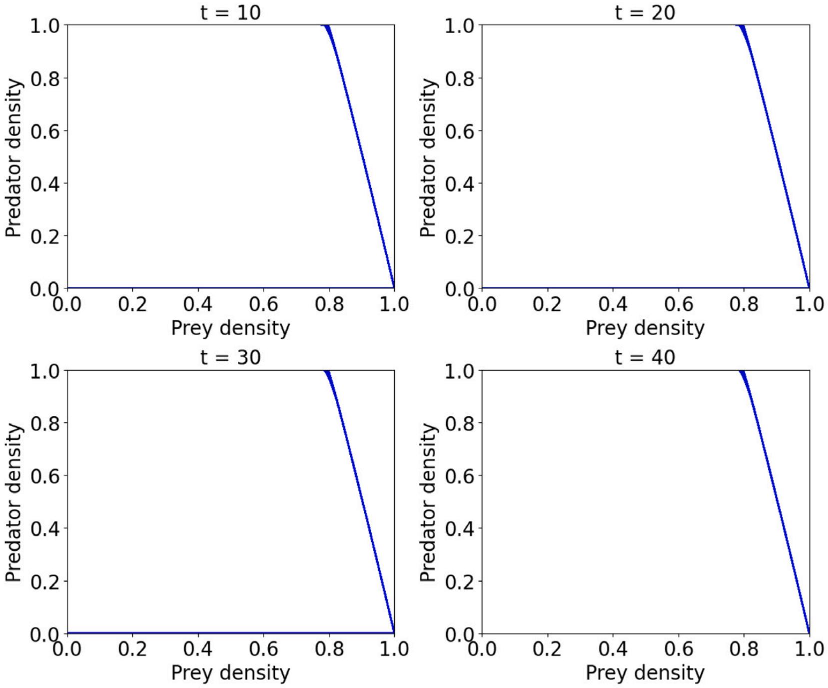
It may be noted that A is the parameter in the prey growth term associated with the Allee effect. Thus, increasing A enhances prey growth under the Allee effect. We observe that as A increases, the prey population front re-forms, and this time, rather than fragmenting into small patches, several layers of the population front emerge, persist, and gradually expand. Nonetheless, within the population front, some small patches also arise, as shown in Fig. 8.

Recall that $B = \frac{n_1}{k}$, which represents the ratio of the Allee threshold n_1 to the environmental carrying capacity k. Reducing B corresponds to diminishing the strength of the Allee effect. In this scenario, we observe that a reduction of B by only 0.02 is sufficient to suppress patchy invasion. Examining the graph (see Fig. 9), we see that a clear prey invasion front forms, and the prey population spreads uniformly along this front. Nevertheless, there are still marked differences compared with Fig. 5, which shows the dynamics in the absence of the Allee effect. The main difference is that in Fig. 5, the central part of the invasion front still maintains relatively low prey density, whereas in the current figure, only the prey density along the invasion front is visible. A plausible explanation is that the Allee effect renders the prey growth rate negative whenever the density falls below B, which in turn leads to reduced prey density in the central region of the invasion front in this case.

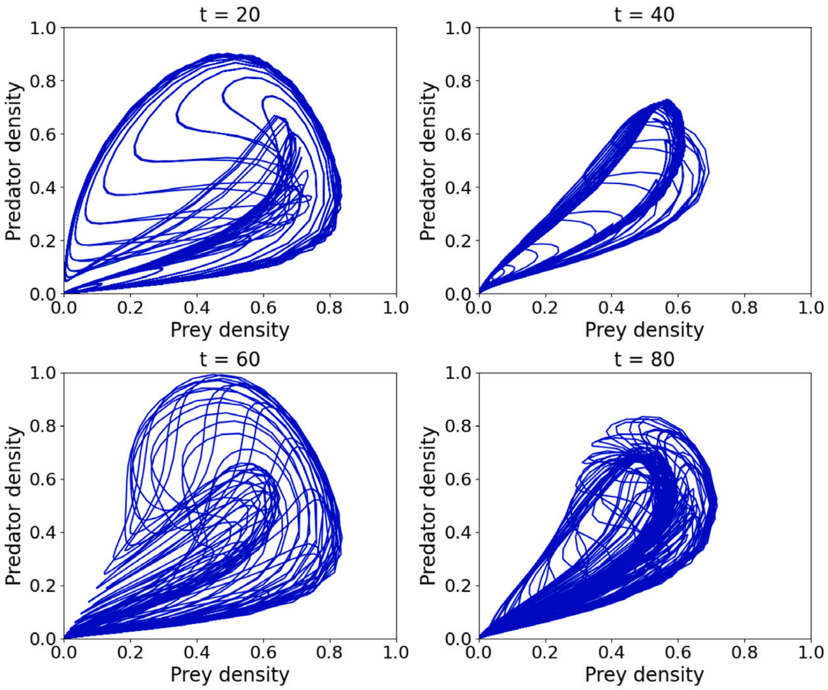
Recall that S denotes prey saturation. Thus, increasing S reduces predator density while increasing prey density. Comparing with Fig. 6 at large times, we observe that higher prey saturation causes the prey to disperse more irregularly, resulting in a greater number of patches. Biologically, this may be explained by the fact that as prey saturation increases, predator density declines and prey density rises, weakening the intensity of the invasion process and allowing the prey to spread in a more scattered manner (see Fig. 10).

5. Case 2: Model dynamics in the absence of predator mortality ($d = 0$)

In this section, we set $d = 0$, which yields $E = \frac{cd}{emk} = 0$. Biologically, this corresponds to assuming that predator mortality is represented by a nonlinear term rather than a linear one, and that the term hy^2 in the model represents a combination of nonlinear mortality and intraspecific competition among predators.



(a)



(b)

Fig. 7. Phase portraits of predator density versus prey density in the presence of the Allee effect. The comparison indicates that without the Allee effect the invasion process produces a relatively uniform phase portrait, whereas the inclusion of the Allee effect leads to irregular and heterogeneous dynamics.

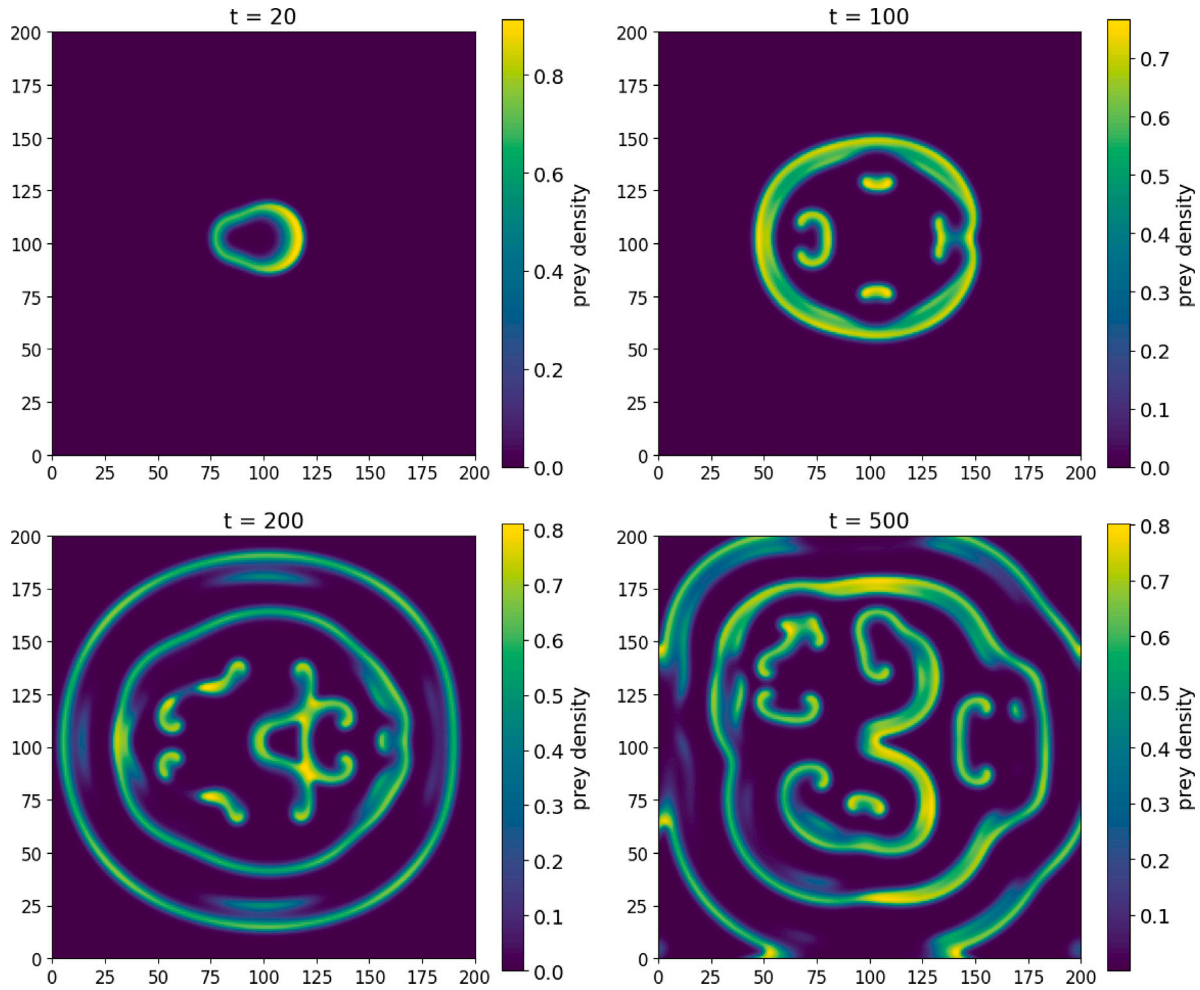


Fig. 8. Spatial distribution of prey density when the parameter A is increased from 4 to 5, while all other parameters and initial conditions remain unchanged.

The non-dimensional model in this case is:

$$\frac{\partial x}{\partial t} = Ax(1-x)(x-B) - \frac{xy}{Sx+1} + \nabla^2 x \tag{14a}$$

$$\frac{\partial y}{\partial t} = \frac{xy}{Sx+1} - Fy^2 + D\nabla^2 y. \tag{14b}$$

After incorporating the nonlinear term into our model, it is no longer possible to obtain a closed-form expression for the positive equilibrium. Consequently, we must specify particular parameter values and use numerical techniques to determine the equilibrium points. This reliance on numerical computation complicates the application of standard stability analysis.

In this section, our primary aim is to provide an illustrative example of Turing pattern formation rather than to undertake a comprehensive analytical investigation. In contrast to Case 1, it is generally not feasible here to derive explicit analytical conditions. To examine the conditions under which Turing instability occurs, we first fix the parameter values and then compute the equilibrium numerically. If a positive equilibrium is obtained, we subsequently test whether it satisfies the analytical Turing instability criterion for those parameter values. For completeness, we restate below the Turing instability condition for our non-dimensional model:

$$\begin{cases} a_{11} + a_{22} < 0, \\ a_{11}a_{22} - a_{21}a_{12} > 0, \\ Da_{11} + a_{22} > 2\sqrt{D}\sqrt{\det(J_{E_4})}. \end{cases} \tag{15}$$

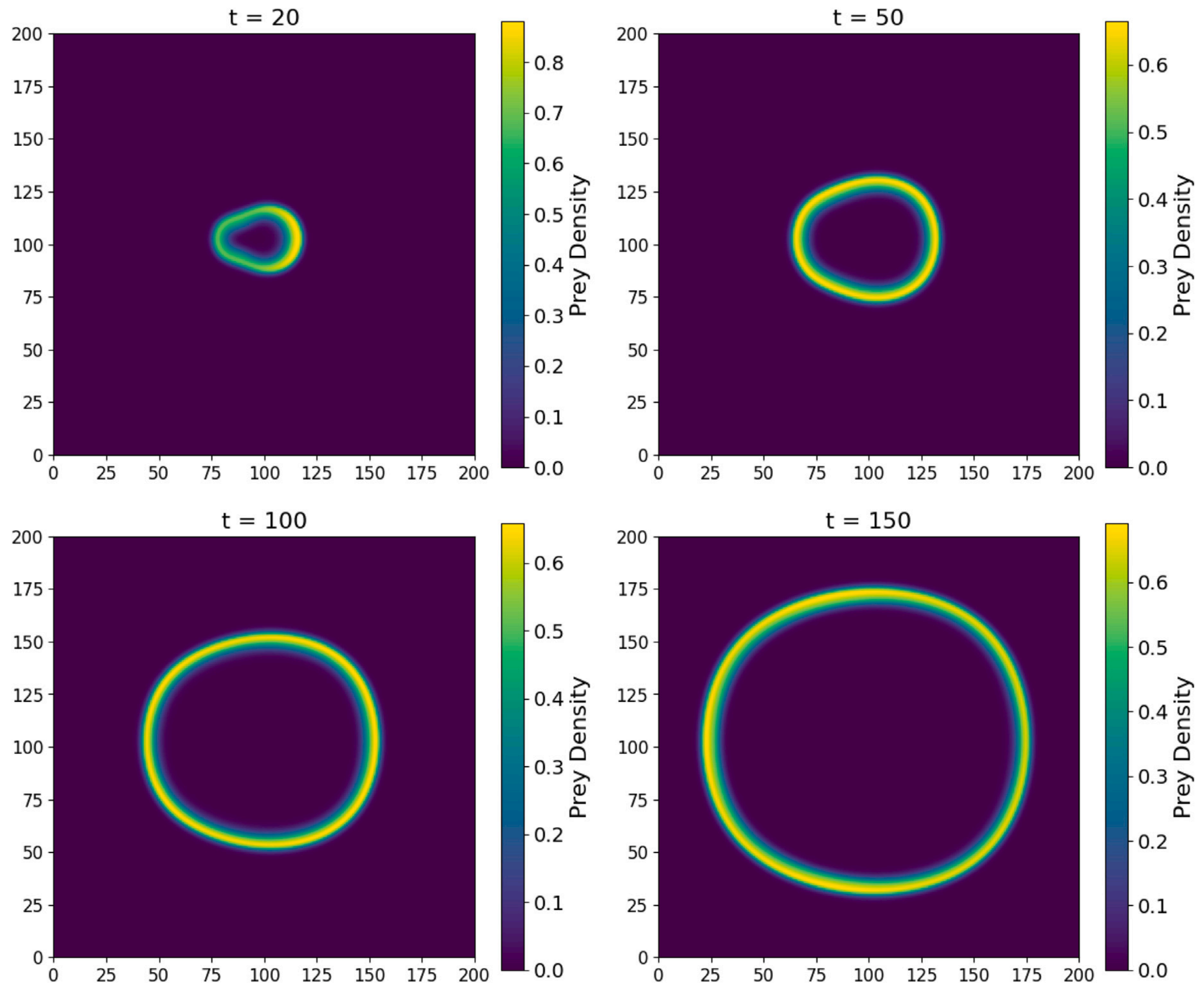


Fig. 9. Spatial distribution of prey density obtained by decreasing the parameter B from 0.22 to 0.20, while all other parameters and initial conditions remain unchanged.

Table 2
Simulation parameters for different cases.

No.	Parameter values	Space step	Time	Figure
1	$A = 1.5,\ B = 0.02,\ S = 0.45,\ F = 0.92,\ D = 60$	10	1000	Fig. 11(a)
2	$A = 2,\ B = 0.01,\ S = 0.45,\ F = 0.6,\ D = 20$	1	1000	Fig. 11(b)
3	$A = 2,\ B = 0.01,\ S = 0.45,\ F = 0.6,\ D = 5$	1	1000	Fig. 11(c)
4	$A = 2,\ B = 0.01,\ S = 0.45,\ F = 0.65,\ D = 21$	1	1000	Fig. 11(d)

We will use the parameter values from [20], ie. ($A = 1.5, B = 0.02, F = 0.92, D = 60$). In order to satisfy the analytical conditions of patterns, We choose $S = 0.45$, which gives equilibrium point $(0.49486929, 0.43993235)$, the three analytical conditions are then:

$$\begin{cases} a_{11} + a_{22} = -0.316742293948303 < 0, \\ a_{11}a_{22} - a_{21}a_{12} = 0.0834888338104531 > 0, \\ Da_{11} + a_{22} - 2\sqrt{D}\sqrt{\det(J^*)} = 0.398683622877765 > 0. \end{cases} \tag{16}$$

We can then plot the resulting spatial patterns, as shown in Fig. 11, using the parameter values listed in Table 2, which are adopted from the literature.

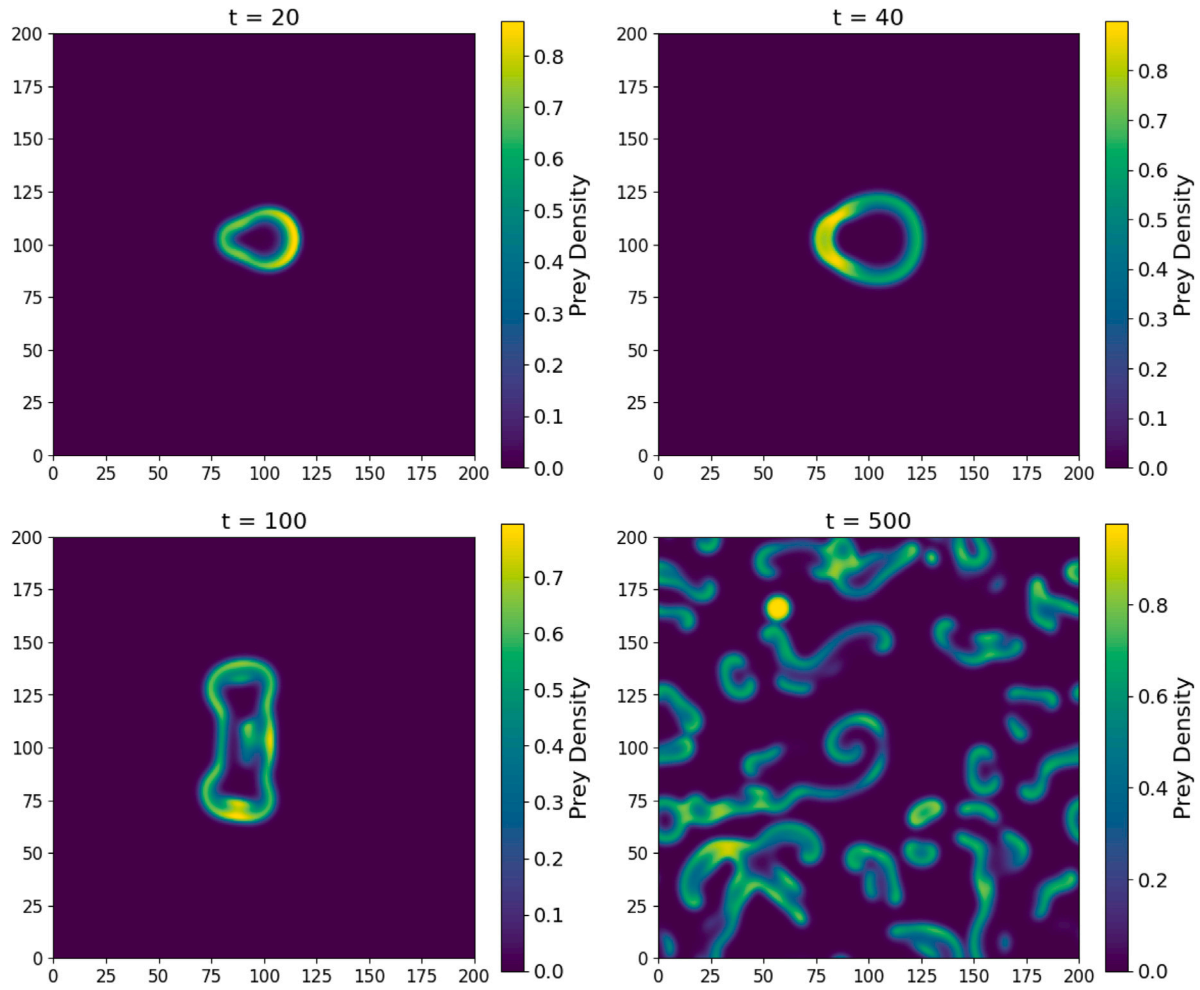


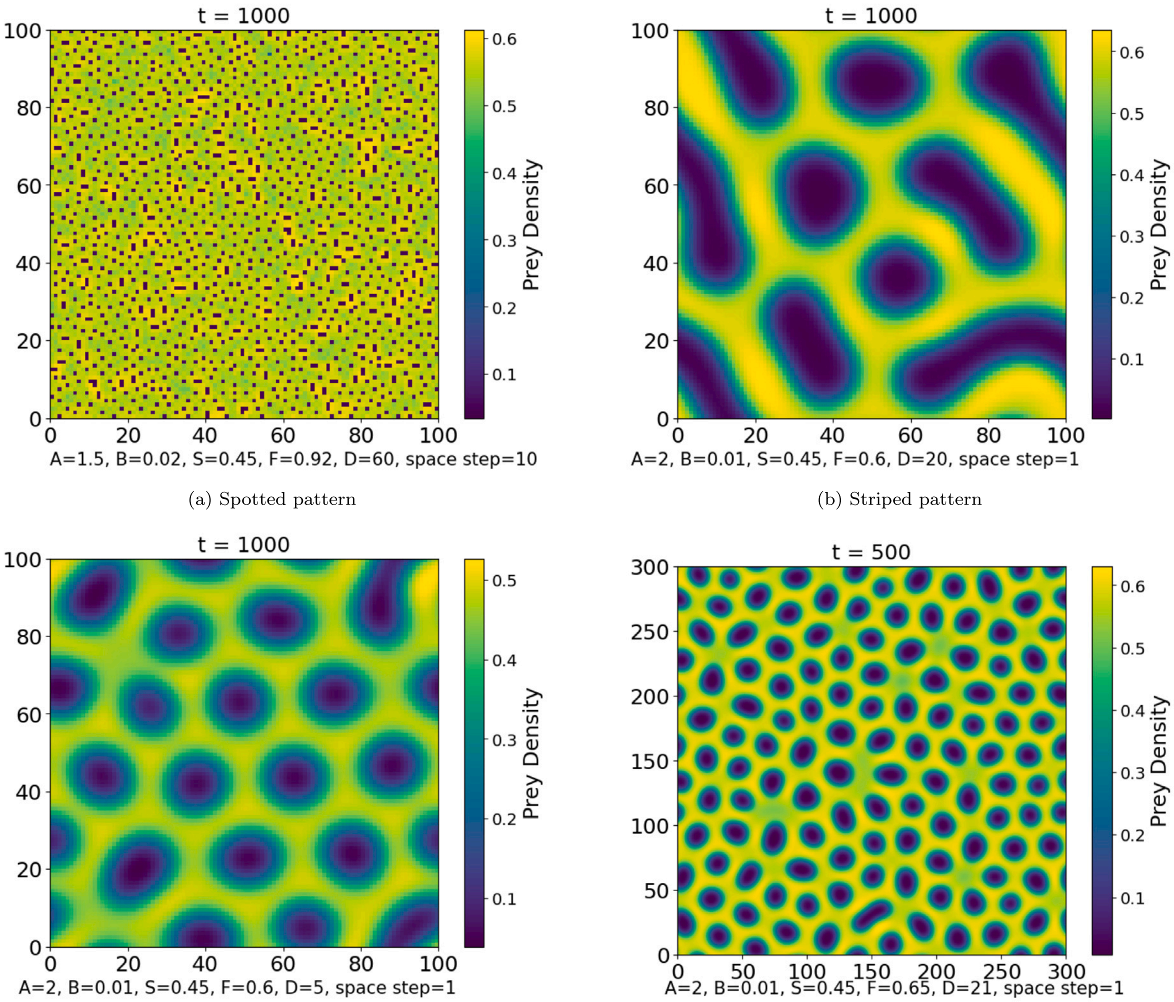
Fig. 10. Spatial distribution of prey density obtained by increasing the parameter S from 0.15 to 0.18, while all other parameters and initial conditions remain unchanged.

The above analysis demonstrates that diffusion-driven instability can arise in the absence of linear predator mortality, leading to the formation of spatially heterogeneous patterns. These results highlight the role of density-dependent effects in shaping system dynamics and indicate that nonlinear interactions alone may be sufficient to generate Turing-type structures under certain parameter regimes. However, from a biological perspective, the assumption $d = 0$, corresponding to the absence of natural predator mortality, limits the ecological realism of this case. Although the nonlinear term hy^2 captures density-dependent mortality and intra-species competition, it does not fully represent the baseline death rate typically observed in real predator populations. As a consequence, the patterns obtained in this regime should be interpreted with caution, as they may arise primarily from the mathematical structure of the model rather than reflecting realistic ecological processes. Therefore, Case 2 should be regarded mainly as a theoretical framework that illustrates how diffusion-driven instability can occur under simplified assumptions. In order to obtain more biologically meaningful insights and to incorporate essential ecological mechanisms, we now turn to a more general formulation of the model. In the following section, we consider Case 3, where both linear predator mortality and density-dependent effects are included, providing a more realistic basis for analyzing system dynamics and pattern formation.

6. Case 3: Analysis of the full model with predator mortality and competition ($d, h \neq 0$)

In this section, we analyze the complete model where both the linear death of predators and intra-species competition are present, i.e., $d, h \neq 0$. Compared with the previous cases, this general model requires a more comprehensive analysis. The non-dimensional system is given by

$$\frac{\partial x}{\partial t} = Ax(1-x)(x-B) - \frac{xy}{Sx+1} + \nabla^2 x, \tag{17a}$$



(a) Spotted pattern

(b) Striped pattern

(c) Dotted pattern

(d) Dotted pattern with larger spatial scale

Fig. 11. Spatial distributions of prey density illustrating different Turing patterns. The horizontal and vertical axes represent the spatial coordinates x and y. Panels (a)–(d) display spotted, striped, and dotted patterns, respectively. These results indicate that the prey population forms heterogeneous spatial structures, with regions of high density separated by areas of low density.

$$\frac{\partial y}{\partial t} = \frac{xy}{Sx+1} - Ey - Fy^2 + D\nabla^2 y. \tag{17b}$$

Setting $y = 0$ yields $x = 0$, $x = 1$, or $x = B$. Hence, the system admits three trivial equilibrium points: $E_1 = (0, 0)$, $E_2 = (1, 0)$, and $E_3 = (B, 0)$. In what follows, we analyze the stability of each equilibrium and discuss their biological interpretations. The stability conditions of extinction equilibrium $E_1(0, 0)$ and predator-free equilibrium E_2 are discussed in Appendix C.

- The Allee threshold equilibrium is given by $E_3 = (B, 0)$ and the Jacobian at E_3 is

$$J_{E_3} = \begin{pmatrix} AB(1-B) & -\frac{B}{BS+1} \\ 0 & \frac{B}{BS+1} - E \end{pmatrix}.$$

The stability conditions are $AB(1-B) + \frac{B}{BS+1} - E < 0$ and $AB(1-B)\left(\frac{B}{BS+1} - E\right) > 0$. Since $A > 0$ and $B < 1$, the second condition requires $\frac{B}{BS+1} - E > 0$, which contradicts the first. Thus, E_3 cannot be locally stable. The eigenvalues are given by $\lambda_1 = AB(1-B) > 0$ and $\lambda_2 = \frac{B}{BS+1} - E$. When $E < \frac{B}{BS+1}$, we have $\lambda_2 > 0$ and the equilibrium E_3 is an unstable node; when $E > \frac{B}{BS+1}$, then $\lambda_2 < 0$ and E_3 becomes an unstable saddle.
From a biological perspective, the point $(B, 0)$ corresponds to the Allee threshold equilibrium. Because $\lambda_1 > 0$, this equilibrium is always unstable: for low predator mortality (small E), it manifests as an unstable node, whereas for high predator mortality

(large E), it turns into an unstable saddle. This captures the inherent difficulty of maintaining the prey population precisely at the Allee threshold in the presence of predators.
- We denote the positive equilibrium by E_4. For this equilibrium, no explicit closed-form representation is available, rendering a fully analytical stability analysis impossible. Given that our primary focus is on spatiotemporal pattern formation, we study the stability and dynamical behavior of E_4 through numerical simulations.

One parameter set that produces patterns is $A = 1.3$, $B = 0.01$, $S = 0.25$, $E = 0.03$, $F = 0.9$, and $D = 10$, with the positive equilibrium (0.32021496, 0.29608954). At this point, the analytical conditions for Turing instability are satisfied:

$$\begin{cases} a_{11} + a_{22} = -0.0923166107516183 < 0, \\ a_{11}a_{22} - a_{21}a_{12} = 0.0288425092436351 > 0, \\ Da_{11} + a_{22} - 2\sqrt{D\ \det(J_{E_4})} = 0.401054687933867 > 0. \end{cases}$$

To investigate pattern formation, we apply a small perturbation around the equilibrium as the initial condition. With a spatial step size of 1 and a time step of 0.01, we perform simulations up to $t = 1000$. Because of these random perturbations, different runs can produce distinct patterns even when the parameter values are identical. In addition, varying the parameter sets may lead to further types of spatial patterns in the model's dynamics. In the following section, we present numerical results obtained from several different parameter combinations.

6.1. Table of parameters in numerical simulation

No.	parameter value	Space step	Time	figure
1	A=1.5,B=0.02,S=0.3,E=0.015,F=0.9,D=20	1	1000	Fig. 12(a)
2	A=1.5,B=0.01,S=0.3,E=0.01,F=0.9,D=20	1	1000	Fig. 12(b)
3	A=1.5,B=0.01,S=0.3,E=0.01,F=0.9,D=60	2	1000	Fig. 12(c)
4	A=1.5,B=0.02,S=0.15,E=0.1,F=0.8,D=30	3	500	Fig. 12(d)
5	A=1.5,B=0.02,S=0.15,E=0.1,F=0.7,D=15	1	500	Fig. 12(e)
6	A=1.3,B=0.01,S=0.28,E=0.02,F=0.9,D=17	1	1000	Fig. 12(f)
7	A=1.35,B=0.05,S=0.45,E=0.12,F=0.85,D=1500	15	3500	Fig. 13(a)
8	A=1.5,B=0.01,S=0.28,E=0.02,F=0.9,D=20	1	6000	Fig. 13(b)
9	A=1.4,B=0.01,S=0.28,E=0.02,F=0.9,D=10	1	1000	Fig. 13(c)
10	A=1.4,B=0.01,S=0.25,E=0.03,F=0.9,D=16	1	1000	Fig. 13(d)
11	A=1.592,B=0.011,S=0.241,E=0.092,F=0.45,D=10	1	1000	Fig. 13(e)
12	A=2.31,B=0.028,S=0.034,E=0.085,F=0.496,D=10	1	1000	Fig. 13(f)
13	A=1.3,B=0.01,S=0.2,E=0.03,F=0.92,D=25	1	100	Fig. 14(a)
14	A=1.25,B=0.01,S=0.15,E=0.1,F=0.9,D=100	3	500	Fig. 14(b)
15	A=1.25,B=0.01,S=0.15,E=0.1,F=0.9,D=175	10	500	Fig. 14(c)
16	A=1.35,B=0.05,S=0.45,E=0.12,F=0.85,D=1100	20	1000	Fig. 14(d)
17	A=1.3,B=0.01,S=0.2,E=0.03,F=0.92,D=20	1	1000	Fig. 14(e)
18	A=1.3,B=0.01,S=0.2,E=0.03,F=0.92,D=10	1	1000	Fig. 14(f)

7. Numerical methods

The spatiotemporal dynamics of the proposed reaction–diffusion system are investigated numerically on a two-dimensional spatial domain. The system is solved using a finite difference scheme for spatial discretization and an explicit time-stepping method for temporal integration. The initial conditions are taken as small random perturbations around the homogeneous steady state with a fixed amplitude ($\epsilon = 0.01$) and uniformly distributed random fluctuations. The spatial domain is discretized into a uniform grid of size $N_x \times N_y$ with spatial step sizes $\Delta x = \Delta y$. The Laplacian operator is approximated using the standard second-order central difference scheme. Zero-flux (Neumann) boundary conditions are imposed on all boundaries of the domain, ensuring that no population flux crosses the boundary. Numerically, these conditions are implemented using reflective (mirror) boundary points. For time integration, an explicit forward Euler method is employed with a fixed time step Δt. The choice of Δt and Δx satisfies the standard stability criterion for diffusion-driven systems, namely

$$\Delta t \le \frac{(\Delta x)^2}{4D_{\max}},$$

where $D_{\max} = \max(D_1, D_2)$ denotes the maximum diffusion coefficient in the system. This condition ensures the numerical stability of the scheme. Simulations are carried out on a square domain of size $L \times L$, with appropriate initial conditions consisting of small random perturbations around the homogeneous steady state to initiate pattern formation. The system is evolved up to a sufficiently large time to observe stable spatial structures. All numerical simulations are performed using uniform parameter settings unless otherwise stated, and the results are verified to be independent of the grid resolution. The numerical parameter values used for the simulations are listed in Section 6.1. The emergence of Turing patterns under these parameter settings can be clearly observed in Figures 12, 13, and 14.

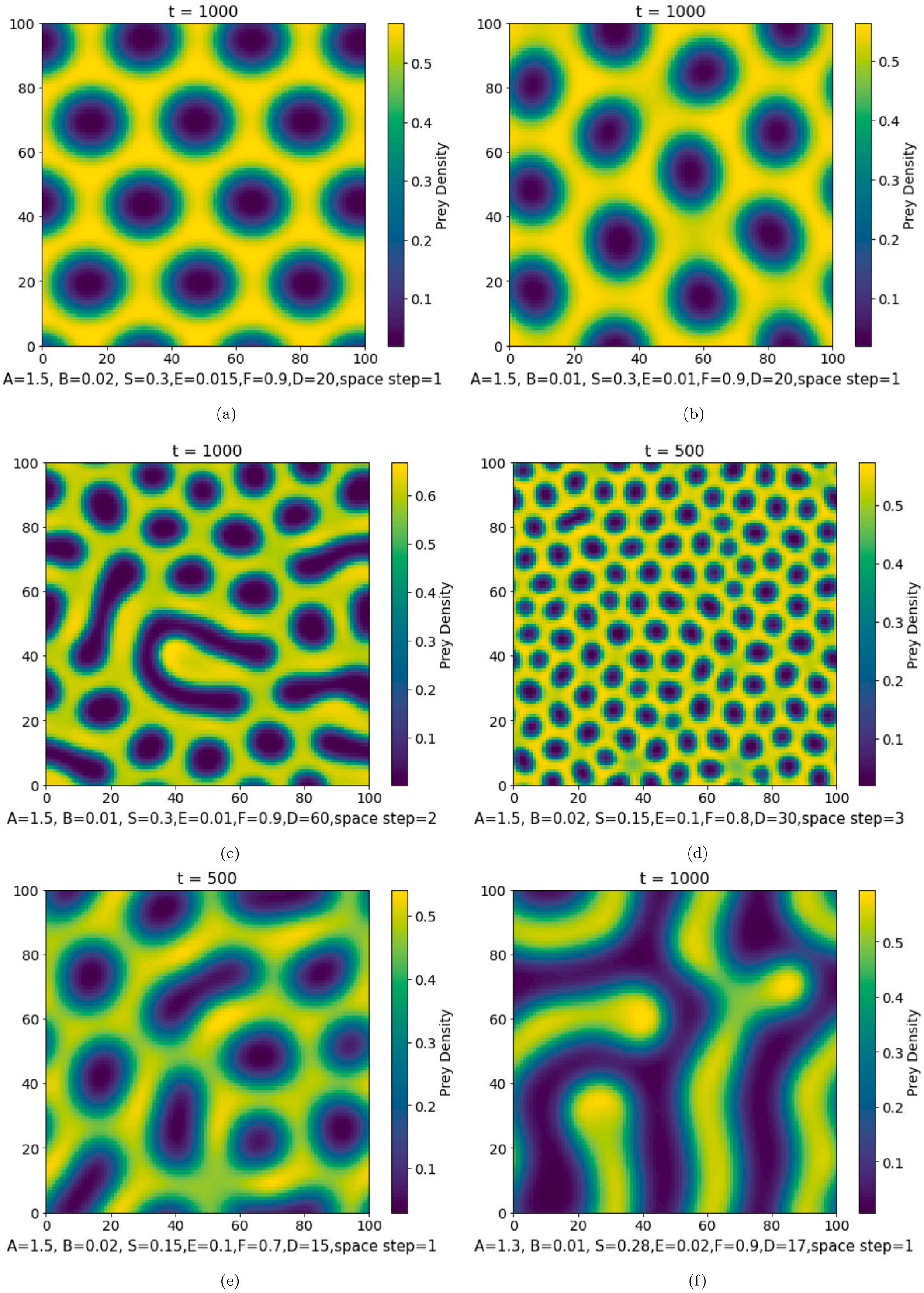


Fig. 12. Spatial distributions of prey density for different parameter configurations. The horizontal and vertical axes represent spatial coordinates, while parameter values are indicated within the panels.

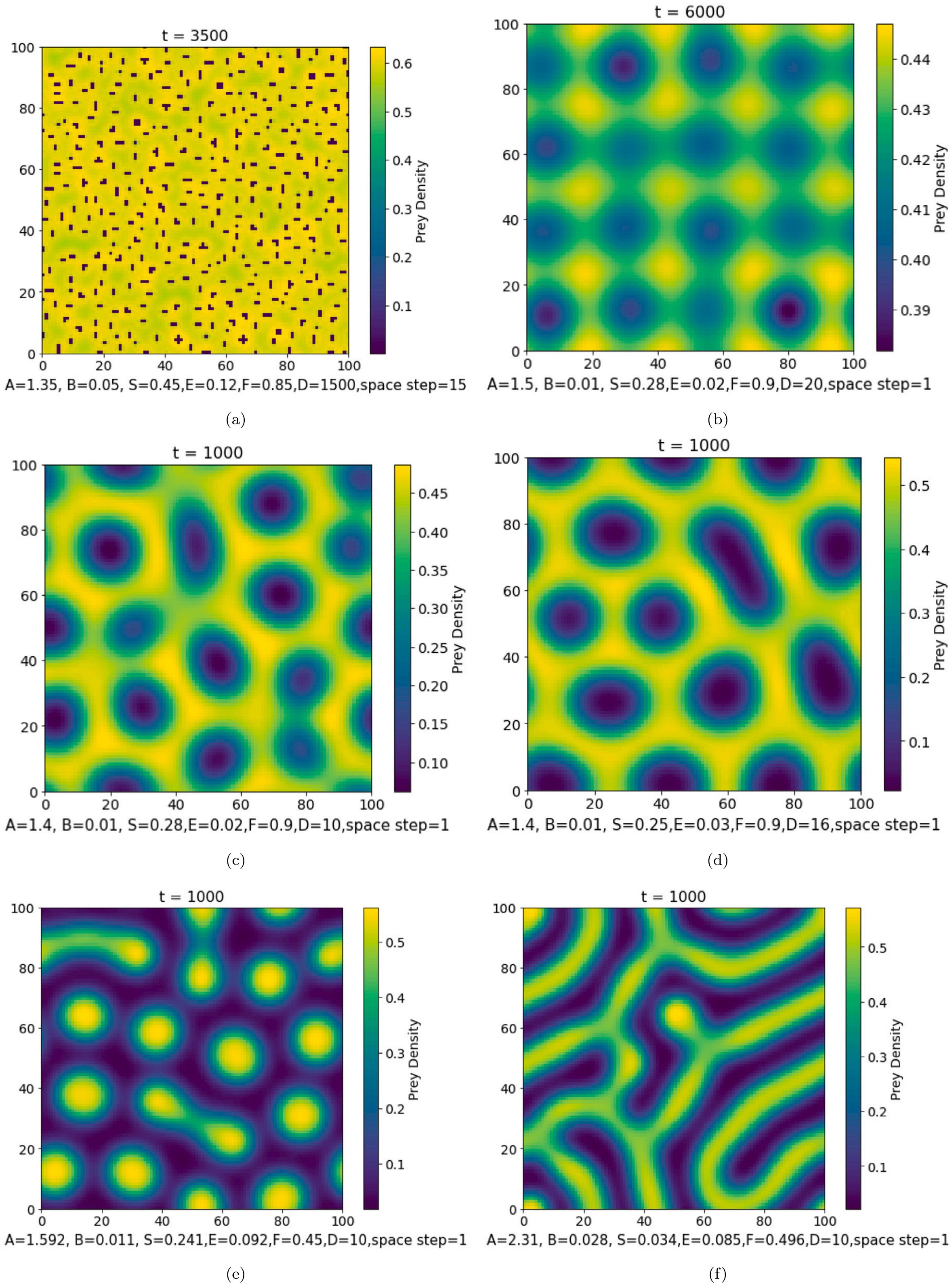


Fig. 13. Spatial distributions of prey density illustrating different pattern morphologies obtained under varying parameter values.

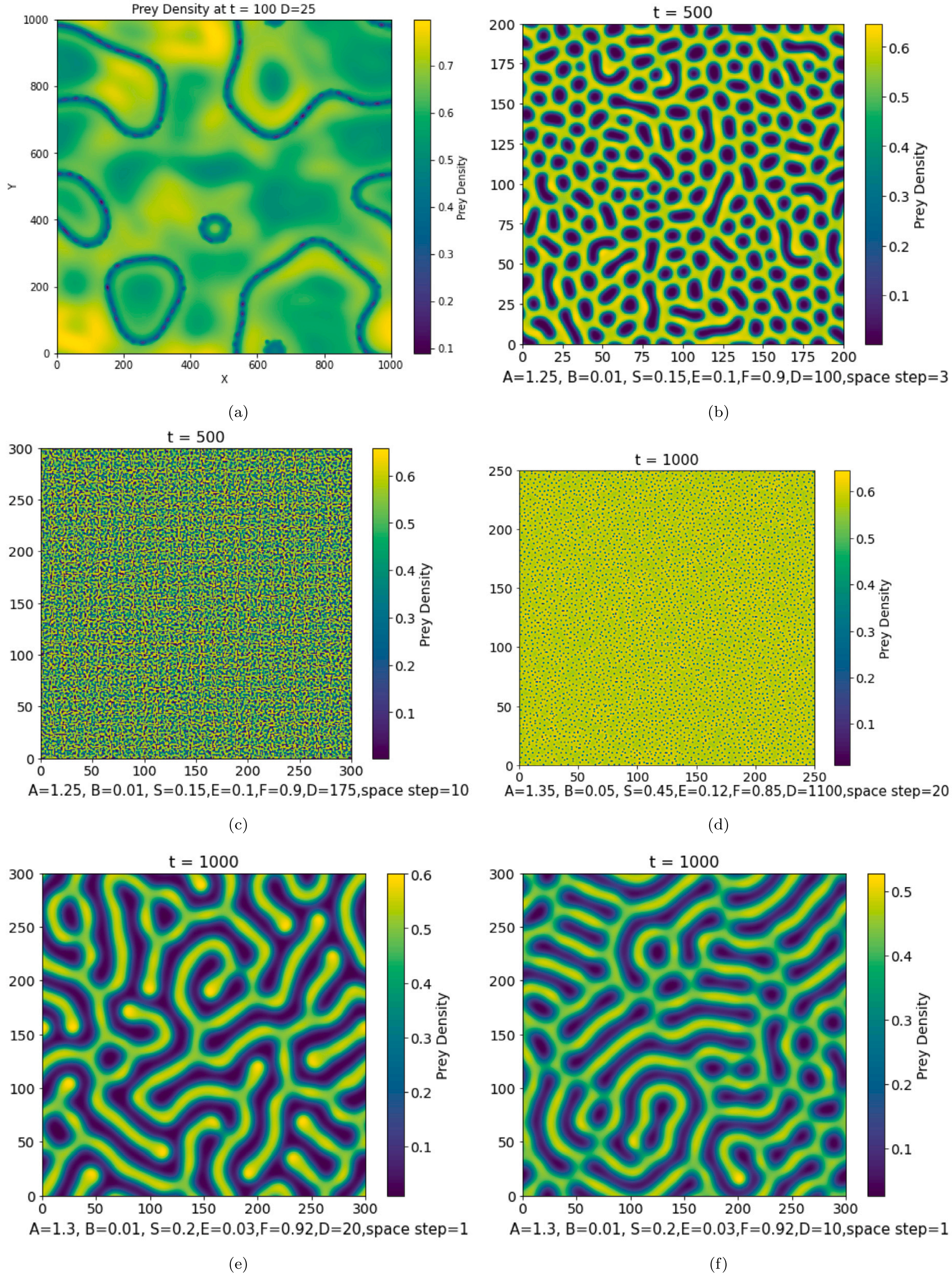


Fig. 14. Spatial patterns obtained in an expanded spatial domain. Parameter values in panel (a): $A = 1.3$, $B = 0.01$, $S = 0.2$, $E = 0.03$, $F = 0.92$, $D = 25$, with step size 1.

8. Network analogue and Laplacian matrix approach to turing patterns

So far, our analysis has been confined to continuous reaction diffusion systems in two-dimensional space. However, many real-world ecological and biological systems are not embedded in continuous media but instead consist of discrete, interacting units such as habitat patches, sub-populations, or individuals linked by dispersal or interaction pathways. This motivates the study of network analogues of reaction–diffusion systems, where the underlying structure is represented as a graph.

In a network setting, each node corresponds to a local population of prey and predators, while edges represent dispersal or migration routes between nodes. Diffusion is no longer described by the spatial Laplacian operator (∇^2), but instead by the graph Laplacian (LA matrix), which encodes the connectivity and topology of the network. This substitution enables us to explore how Turing instabilities and pattern formation are shaped not just by biological parameters but also by network architecture. The specific contribution of the present work, relative to prior studies on network Turing instabilities, is to examine how three biologically motivated features of the proposed model the strong Allee effect, a saturating prey-dependent functional response, and predator intra-species competition collectively shape pattern formation across networks of varying connectivity. By applying the same parameter sets validated in Section 6 to lattice networks of varying degree, we demonstrate that network connectivity governs which of the pattern morphologies admitted by these ecological mechanisms is ultimately realized.

8.1. Turing patterns on networks

When reaction–diffusion dynamics are projected onto a network, uniform equilibria can lose stability due to the interplay between reaction kinetics and heterogeneous diffusion across the network. This process, known as a network Turing instability, generates stationary but heterogeneous patterns in the distribution of prey and predators over the nodes. Unlike the smooth spatial patterns observed in continuous domains (e.g., spots, stripes, labyrinths), network-based Turing patterns reflect the structural properties of the underlying graph. The type of pattern that emerges depends strongly on features such as node degree distribution, modularity, clustering, or small-world effects.

Mathematically, the dynamics on a network of N nodes can be written as:

$$\frac{dx_i}{dt} = f(x_i, y_i) + D_x \sum_{j=1}^{N} L_{ij} x_j, \quad \frac{dy_i}{dt} = g(x_i, y_i) + D_y \sum_{j=1}^{N} L_{ij} y_j, \tag{18}$$

where x_i and y_i denote prey and predator densities at node i, f, g represent the local reaction terms, D_x, D_y are diffusion coefficients, and L is the graph Laplacian matrix.

8.2. Linear stability and dispersion relation on networks

Linearizing system (18) around the homogeneous equilibrium (x^*, y^*) and expanding perturbations in the Laplacian eigenbasis (see Appendix D for the full derivation) yields the network dispersion relation

$$\lambda^2 - \lambda\left[\mathrm{tr}(J) - \Lambda_k(D_x + D_y)\right] + \det(J - \Lambda_k \mathcal{D}) = 0, \tag{19}$$

where $\mathcal{D} = \mathrm{diag}(D_x, D_y)$ and Λ_k is the kth eigenvalue of the graph Laplacian. A Turing instability on the network occurs if:

1. The homogeneous equilibrium is stable without diffusion: $\mathrm{tr}(J) < 0$, $\det(J) > 0$.
2. There exists at least one nonzero Laplacian eigenvalue $\Lambda_k > 0$ such that $\mathrm{Re}(\lambda(\Lambda_k)) > 0$.

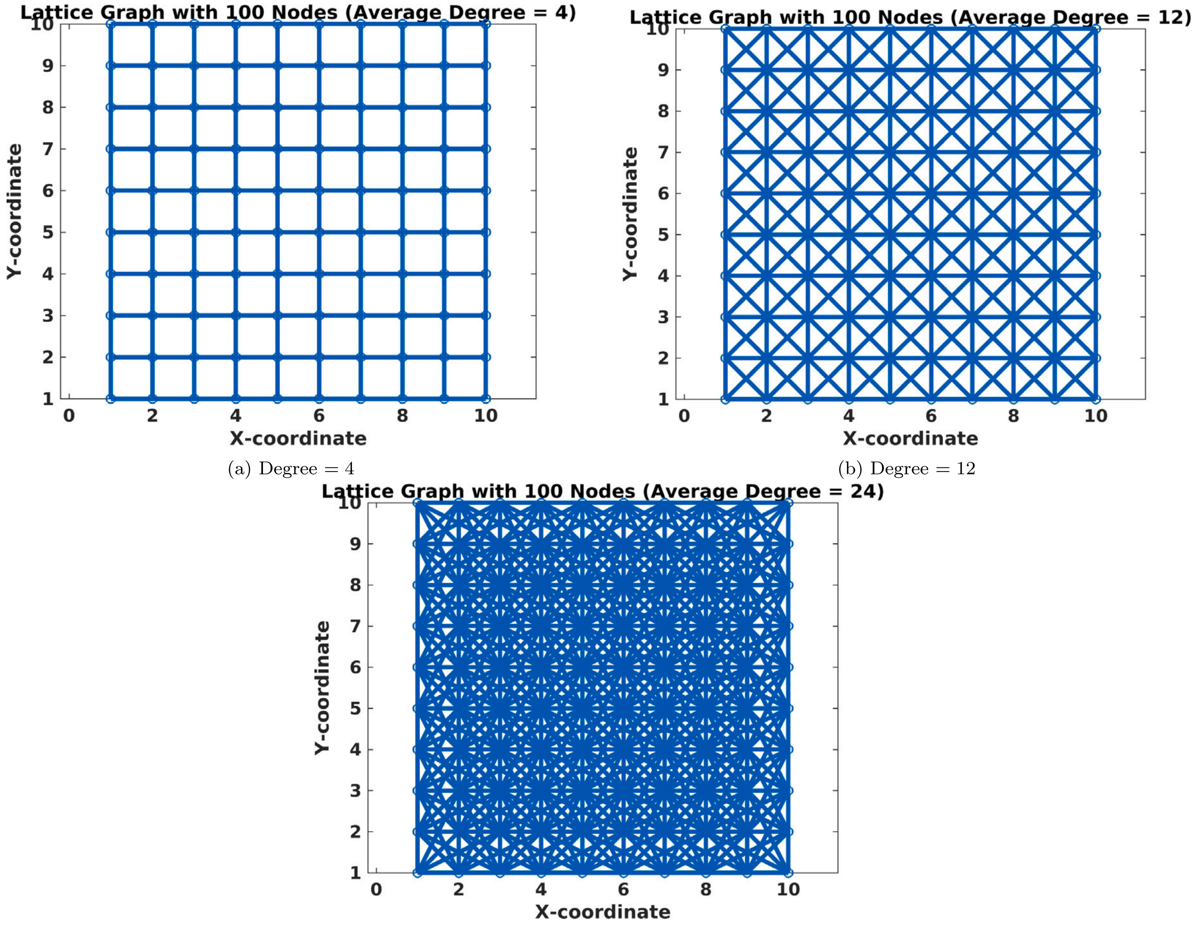
Thus, the onset of Turing instability depends not only on the reaction parameters but also on the Laplacian spectrum of the underlying network. Different network topologies, having different eigenvalue spectra, may therefore either permit or suppress pattern formation under identical reaction parameters.

8.3. The Laplacian matrix (LA matrix)

In graph theory, the combinatorial Laplacian is commonly defined as $L = D - A$. In this work, we adopt the alternative convention $L = A - D$, which differs by an overall negative sign. This formulation has also been used in several studies of diffusion dynamics on networks [6,8,21], where the diffusion term is written with a positive sign in the dynamical equations. In the matrix, cA is the adjacency matrix ($A_{ij} = 1$ if there is a link between node i and node j, otherwise 0) and d is the diagonal degree matrix with entries $d_{ii} = \sum_j A_{ij}$. By construction, each row of L sums to zero, reflecting conservation of mass under diffusion.

The eigenvalues and eigenvectors of L play a central role in pattern formation. When linearizing the system around a uniform steady state, perturbations decompose into eigenmodes of the Laplacian. For each eigenvalue λ_k, the stability condition depends on the balance between reaction terms and diffusion terms scaled by λ_k. A Turing instability occurs if some eigenmodes grow (positive real part of growth rate), leading to non-uniform distributions of prey and predator densities across the network.

To illustrate the effect of connectivity on the Laplacian spectrum, we computed the eigenvalue distributions for lattice networks with different average degrees. Fig. 15 shows three representative cases: (a) average degree = 4, (b) average degree = 12, and (c) average degree = 24. As the average degree increases, the Laplacian spectrum becomes broader, indicating stronger diffusion coupling across the network. Since Turing instabilities are closely tied to the eigenvalues of the Laplacian, these results highlight how denser connectivity can shift the onset of instability and alter the type of emergent patterns. In sparse networks, instabilities are more localized, while in highly connected networks, they spread more uniformly across nodes.



(a) Degree = 4 (b) Degree = 12

(c) Degree = 24

Fig. 15. Laplacian spectra for lattice networks with different average degrees. Panels (a)–(c) correspond to average degrees 4, 12, and 24, respectively.

8.4. Implications of the Laplacian framework

This Laplacian-based approach extends the classical Turing theory to discrete and irregular topologies. It demonstrates that pattern formation is not solely dependent on biological parameters but also critically on the structure of interactions encoded in the network. For example, networks with strong modularity may localize patterns within clusters, while highly connected networks may suppress spatial heterogeneity. Thus, by studying Turing instabilities through the Laplacian matrix, we gain insights into how ecological or epidemiological processes unfold over real-world landscapes that are inherently discrete and structured. To further investigate how network connectivity influences the emergence of Turing patterns, we simulated the reaction–diffusion dynamics on lattice networks with average degrees 4, 12, and 24 using the same parameter set as in Fig. 14(e). The results are shown in Fig. 16. For the sparse lattice with degree 4, the system exhibits irregular labyrinth-like structures and localized patches, indicating that limited connectivity enhances spatial heterogeneity and produces fragmented patterns. At an intermediate degree of 12, the patterns become more ordered, forming stripe-like structures that reflect a balance between local reaction dynamics and intermediate diffusion coupling. For the highly connected case with degree 24, the strong diffusion suppresses fine-scale heterogeneity, leading to smoother gradients and almost homogeneous distributions. These results confirm that the type of Turing pattern is strongly dependent on network connectivity: sparse networks promote irregular and localized structures, while dense networks tend to suppress heterogeneity and favor smoother, large-scale modes.

The influence of network connectivity on Turing pattern formation is evident from Figs. 17–19, which display the prey density distributions corresponding to the six parameter sets of Figures 2a-2f on lattices with average degrees 4, 12, and 24. For the sparse lattice with degree 4 (Fig. 17), the system supports well-defined spot and labyrinthine patterns, reflecting strong spatial heterogeneity driven by localized diffusion. When the connectivity increases to degree 12 (Fig. 18), the patterns become coarser and more stripe-like, with a reduction in the number of distinct spots. In the highly connected lattice with degree 24 (Fig. 19), the strong diffusive coupling suppresses fine-scale structure, and the resulting configurations approach smooth gradients rather than classical Turing morphologies. These results demonstrate that while parameter values determine the potential for instability, the eventual type

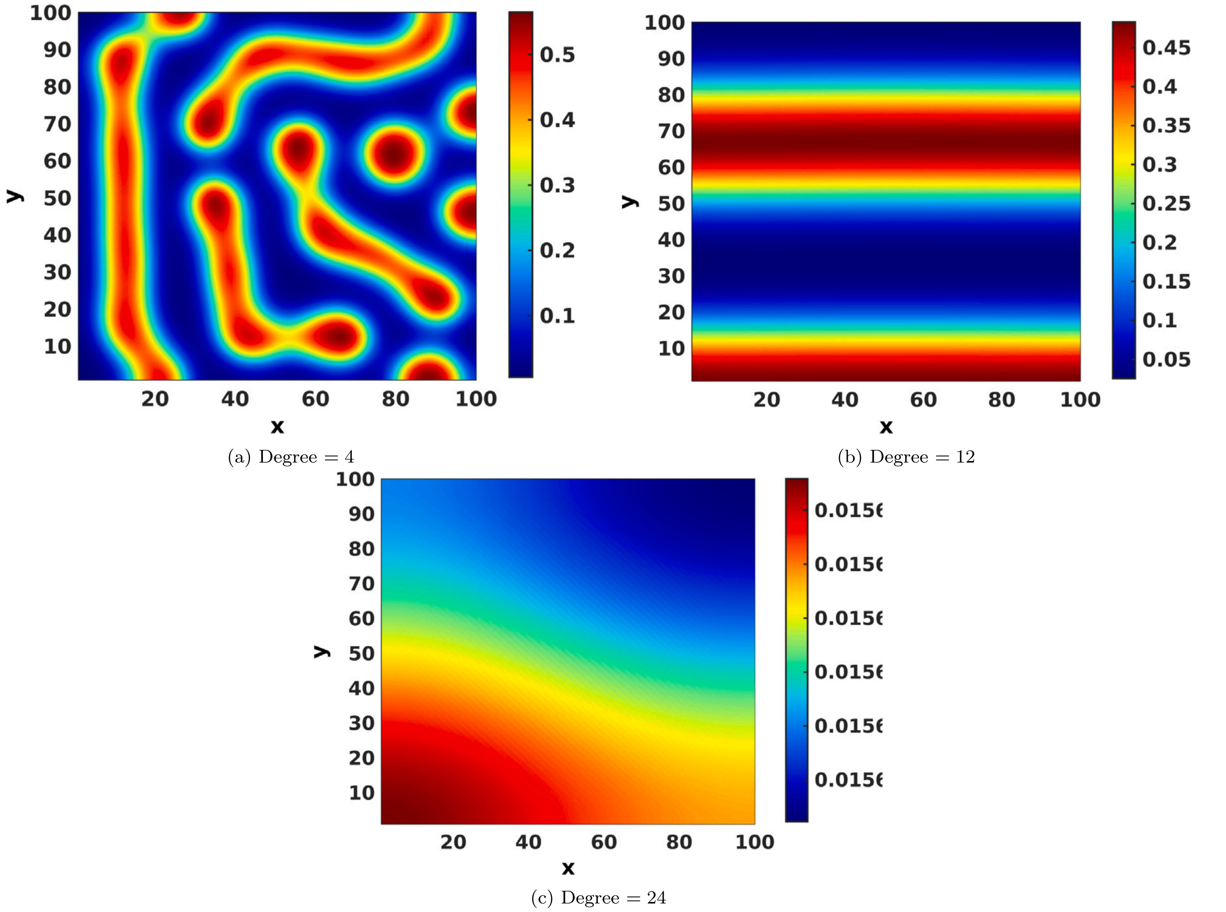


Fig. 16. Laplacian spectra for lattice networks with different average degrees. Panels (a)–(c) correspond to average degrees 4, 12, and 24, respectively.

and scale of emergent patterns are strongly controlled by network connectivity: sparse networks promote irregular spot-labyrinth mixtures, intermediate connectivity yields stripe-like structures, and dense networks homogenize the dynamics, erasing small-scale heterogeneity.

9. Conclusion

In this final section, we will give the conclusion as well as the biological interpretation of our numerical simulation with respect to both the patchy invasion demonstrated in case 1 and the Turing pattern in cases 2 and 3.

9.1. Biological interpretation of patchy invasion

Patchy invasion should not be considered a result caused by any single mechanism. Instead, it is the consequence of a combination of various biological factors [17]. In addition, Patchy invasion is also not driven by Turing instability and is not due to stochastic factors under a deterministic model [18].

In our simulation, we have shown that when there is no Allee effect, prey spreads following a uniform circular population front. However, after we include the Allee effect, the population front breaks into small patches. Irregular geographic distribution of prey begins to form. The reason behind this is that in 2-dimensional space, the prey may be able to escape from the predator at a certain spot, which would result in separate patches in prey distribution. Then, each patchy can begin to expand according to the population front and during this process, some prey may manage to escape again, thus leading to new patches forming [18]. Repeating this patchy formation process, we get the irregular geographic distribution as shown in the simulation part. One important application of patchy invasion depends on whether it demonstrates spatiotemporal chaotic dynamics [17]. This is discussed, and chaos behaviors are observed from [22]. Therefore, it would not be practical to predict the geographic location of patches in a long time period.

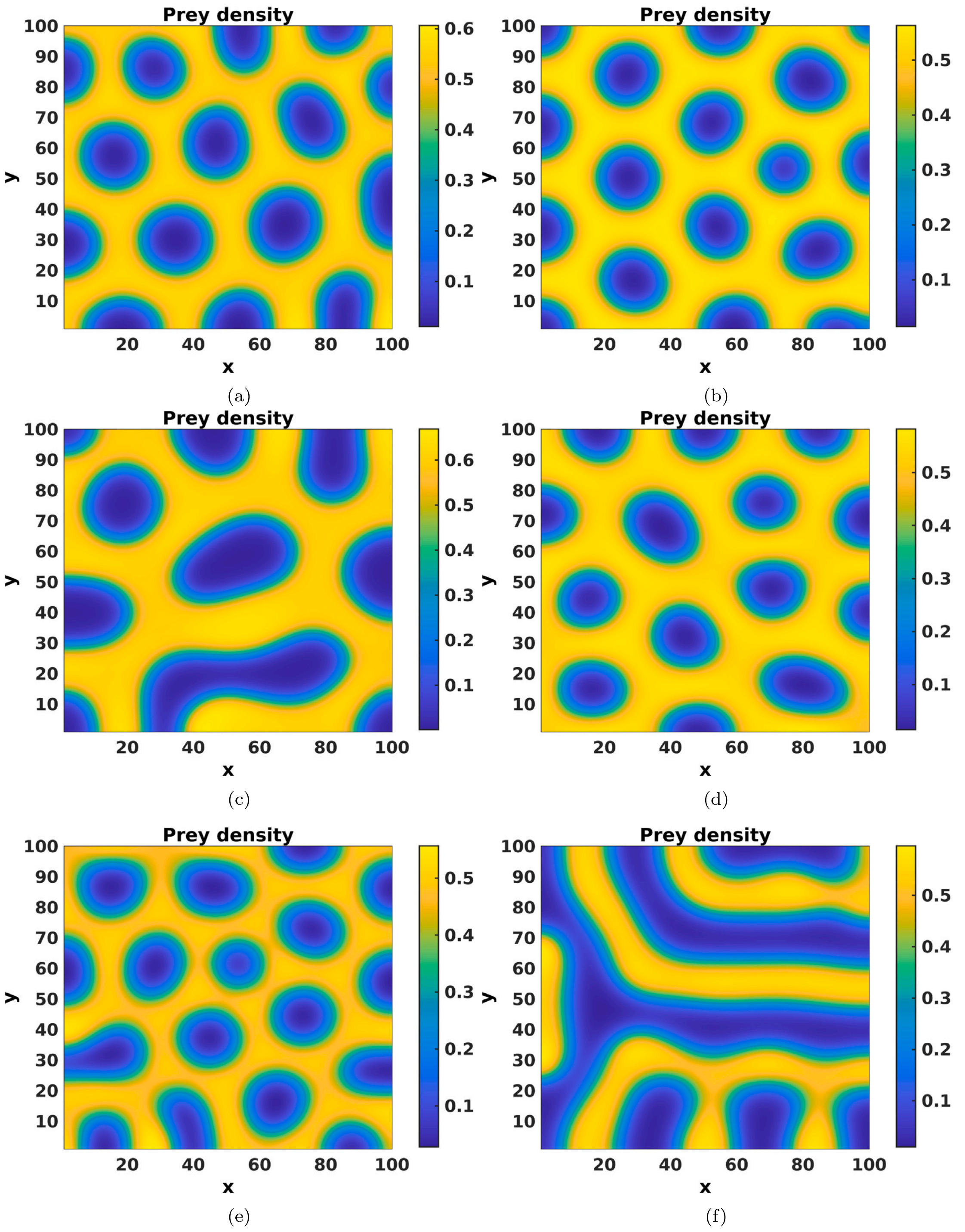


Fig. 17. Turing patterns obtained on a lattice network with average degree 4 for different parameter sets corresponding to Fig. 13.

9.2. Biological interpretation of turing pattern

Turing pattern is a very general topic and can have an enormous influence on a variety of fields. Scientists have studied patterns in fish skin [23], the potential effect of temperature on patterns [24], and the effect of dormancy on patterns [25], which all produce

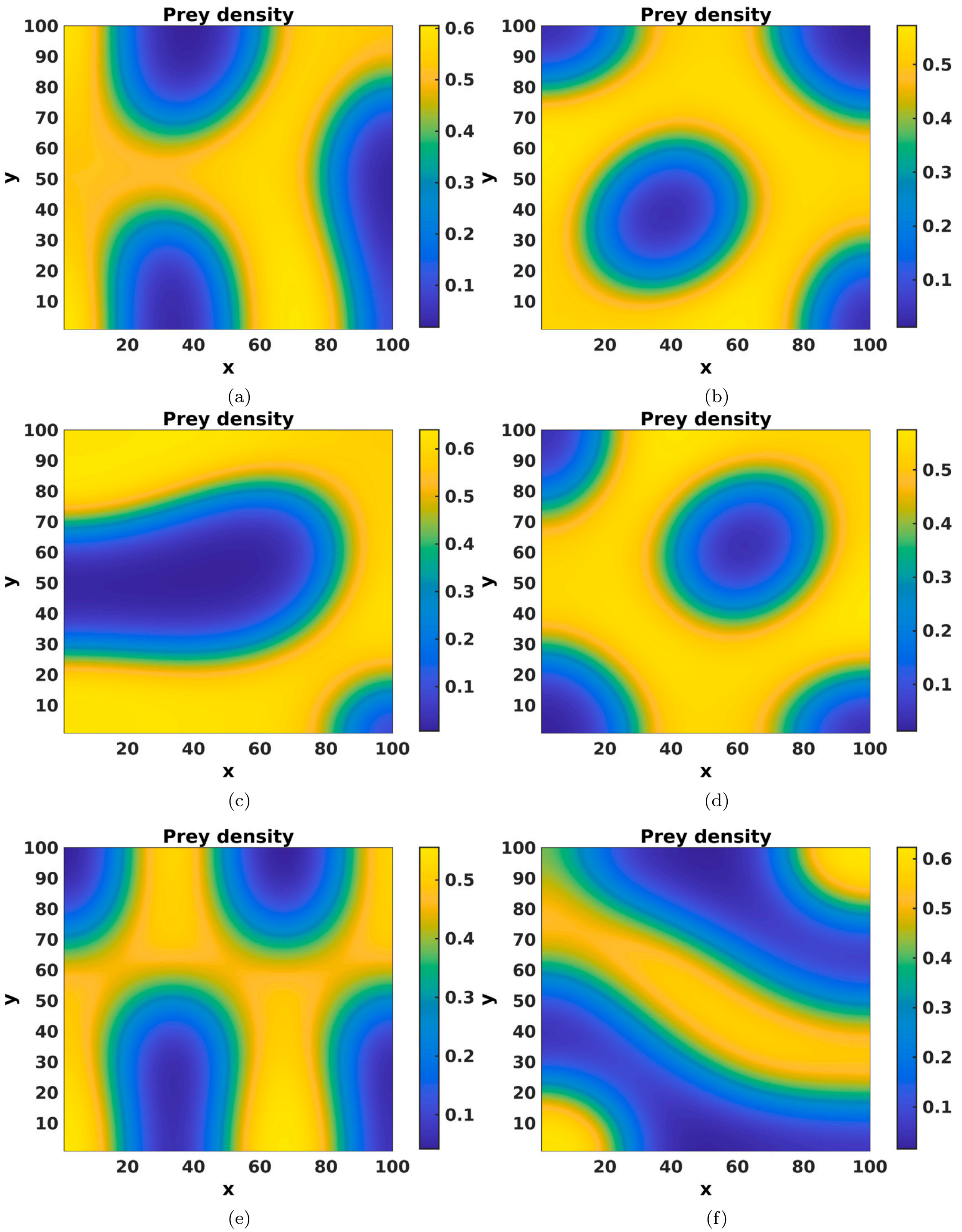


Fig. 18. Turing patterns obtained on a lattice network with average degree 12 for different parameter sets corresponding to Fig. 13.

significant biological meanings. In this paper, we investigate the pattern formation under the prey–predator reaction–diffusion model with Allee effect.

In case 3, We first performed stability analysis on each equilibrium point. To be specific, the extinction point(0,0) is always locally stable due to the Allee effect we included in our model. On the other hand, for (1,0) to be stable, which means that for the predator to die out, a minimum value of death rate of the predator must be exceeded, which is biologically meaningful as a

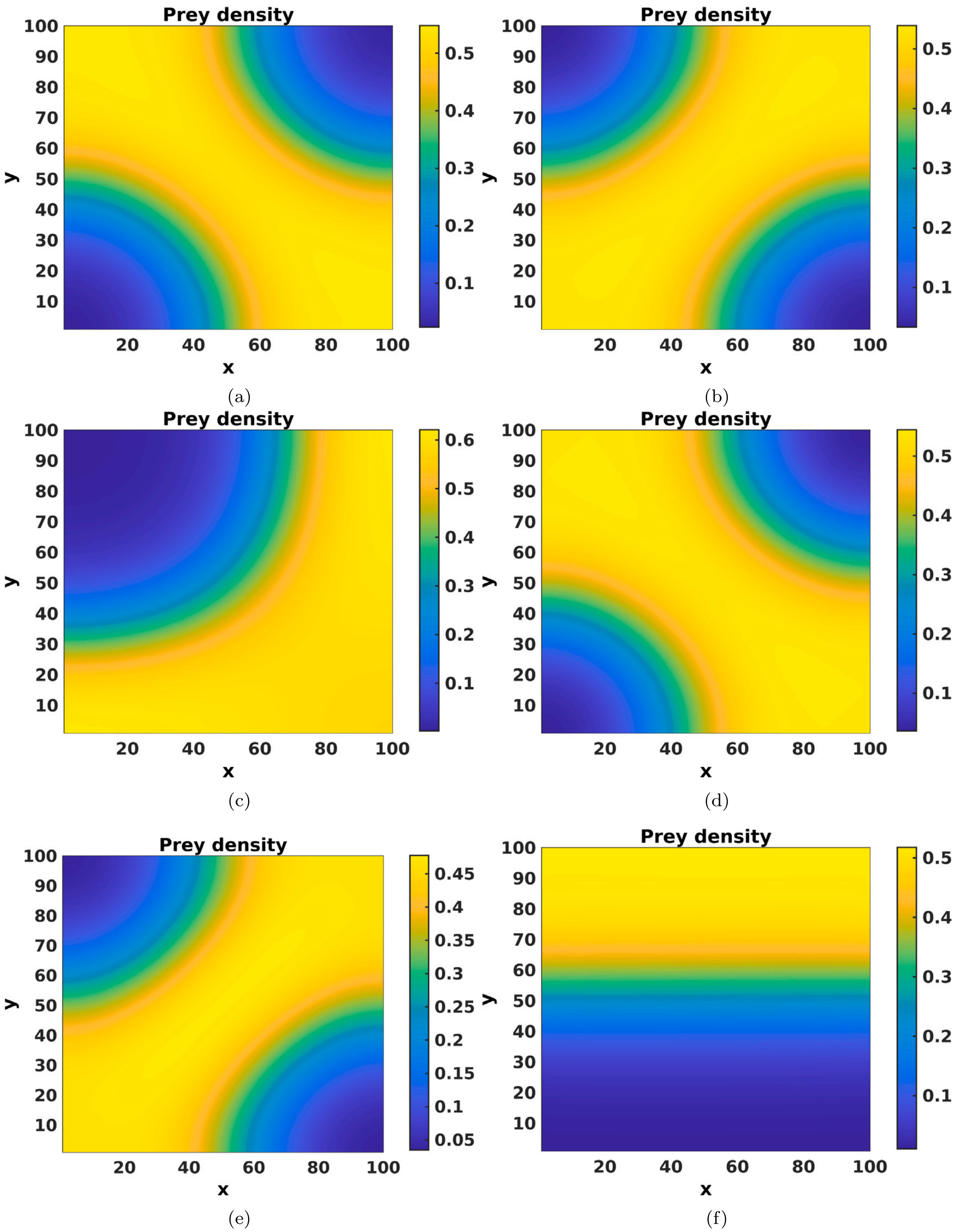


Fig. 19. Turing patterns obtained on a lattice network with average degree 24 for different parameter sets corresponding to Fig. 13.

higher death rate is more likely to lead to extinction. In the case for (B,0), we found that when the death rate of the predator is high enough, this point becomes a saddle point, thus first attracting predator density to a low level and then pushing it away. On the contrary, if the death rate is low, this point is an unstable node, which means that the extinction of the predator would not happen.

Furthermore, by extending the analysis to network-structured domains, we observe that connectivity also plays a decisive role in shaping biological outcomes. For lattices with low average degree (LA4), the prey population forms fragmented spots and labyrinth-like patches, which may be interpreted as localized clusters in sparsely connected landscapes. At intermediate connectivity (LA12), the system exhibits more regular stripe-like structures, suggesting semi-ordered spatial organization as dispersal links increase. In contrast, high connectivity (LA24) suppresses fine-scale variation, leading to almost homogeneous distributions, which biologically correspond to well-mixed populations with weak spatial differentiation. These results emphasize that the network context can either reinforce or diminish spatial self-organization, depending on the degree of connectivity.

Along with the stability analysis, we have also successfully shown a variety of patterns using numerical simulation. From these patterns, we observe that the geographic distribution of prey in 2D space under our model displays dotted, striped, spotted, labyrinth, or combined patterns. This can provide insights into biological population dynamics and geographic distribution. To be specific, the diffusion of prey and predator does not always lead to spatial homogeneous distribution of population density. Especially in the case that includes Allee effect, prey may form irregular patterns and spread non-uniformly.

Turing pattern generally has a much deeper implication in biology. If we look at the general picture of biology, Turing pattern is also suggested to underpin the foundation of the biological embryogenesis process, the skin pattern of vertebrate and vertebrate limb development mechanism [26].

9.3. Summary and future research

In this paper, we have investigated the possible formation of patterns under the prey–predator reaction–diffusion model. In the introduction part, we derived the general model step by step from the beginning and explain the effect and biological interpretation of each of the term. Then, we performed non-dimensionalization on our model for the convenience of numerical simulations. In the following part, we look at the Allee effect in detail and illustrate the significance it possesses in our general model. Next, we put restraints on the parameter values of our model and discuss three separate cases. In the first case where $h = 0$, which means that no intra-species competition is considered in the model, we did stability analysis, proved the existence of Hopf bifurcation, and demonstrated it in two different types of graphs. Most important of all, we managed to prove that the Turing pattern does not exist in this model since the analytical conditions of Turing instability can never be satisfied. In the second case where we set $d = 0$, we performed a similar analysis as in the first case. Furthermore, we found that the analytical conditions for Turing patterns can be satisfied this time and we have used sets of parameters from other papers to show different types of patterns through numerical simulations. However, this case merely serves as an introductory purpose for the third case, which is the general case where we get rid of all the restraints. In this case, We performed a complete analysis of each equilibrium point and gave biological meanings correspondingly. Then we ran a series of numerical simulations and plotted a variety of patterns driven by Turing instability. In the end, we included a biological interpretation section to give real-life meanings to all of our numerical simulations and illustrated the significant role that Turing pattern plays in modern biology.

In addition to the continuous reaction–diffusion framework, we also explored Turing instabilities on network-structured domains by replacing the spatial Laplacian with the graph Laplacian. Our simulations (Figs. 17–19) demonstrate that network connectivity strongly influences the type and scale of emergent patterns: sparse lattices (average degree 4) produce irregular spot-labyrinth mixtures, intermediate connectivity (average degree 12) favors stripe-like configurations, and highly connected lattices (average degree 24) suppress fine-scale heterogeneity, yielding smooth gradients. These findings highlight that the onset and structure of Turing patterns are shaped not only by biological parameters but also by the topology of the underlying interaction network.

We have investigated plenty of population dynamics in our model. However, there is still some advanced work that could be done to dive further into the topic. First, we could try and apply another type of Allee effect and discuss the difference between different types of Allee effect regarding the formation of Turing patterns. In addition, we could set other restraints to the general model and discuss the results under these new models. We have analyzed the model in 2D space and in order to further explore the Turing patterns, we could also do an analysis of pattern formation in 3D space [27]. which would yield completely different graphs and possibly display more biological implications

CRediT authorship contribution statement

Mainul Haque: Writing – review & editing, Visualization, Validation, Supervision. **Junhao Chen:** Methodology, Conceptualization. **Tong Bi:** Methodology, Formal analysis. **Hongxi Chen:** Software, Data curation. **Jiawei Chen:** Software, Investigation, Conceptualization. **Yibin Wang:** Writing – original draft, Formal analysis. **Masoom Bhargava:** Writing – review & editing, Software. **Balram Dubey:** Writing – review & editing, Validation, Software. **Anal Chatterjee:** Writing – review & editing, Visualization.

Declaration of competing interest

The author declares that there is no conflict of interest.

Appendix A. Expressions of the coefficients $a_{ij}, i, j = 1, 2$.

$$\begin{aligned}
a_{11} &= -\frac{AES(ES+E-1)(BES-B+E)}{(ES-1)^4\left(-\frac{ES}{ES-1}+1\right)^2} + \frac{AE\left(-B-\frac{E}{ES-1}\right)}{ES-1} \\
&\quad - \frac{AE\left(\frac{E}{ES-1}+1\right)}{ES-1} + A\left(-B-\frac{E}{ES-1}\right)\left(\frac{E}{ES-1}+1\right) \\
&\quad - \frac{A(ES+E-1)(BES-B+E)}{(ES-1)^3\left(-\frac{ES}{ES-1}+1\right)}, \\
a_{12} &= \frac{E}{(ES-1)\left(-\frac{ES}{ES-1}+1\right)}, \\
a_{21} &= \frac{AES(ES+E-1)(BES-B+E)}{(ES-1)^4\left(-\frac{ES}{ES-1}+1\right)^2} + \frac{A(ES+E-1)(BES-B+E)}{(ES-1)^3\left(-\frac{ES}{ES-1}+1\right)}, \\
a_{22} &= -E - \frac{E}{(ES-1)\left(-\frac{ES}{ES-1}+1\right)}.
\end{aligned}$$

Appendix B. Proof of Theorem 1

The analytical criteria for Turing instability in (13) consist of four conditions. The first condition guarantees the positivity of the equilibrium. The second and third conditions together ensure that the equilibrium is locally asymptotically stable in the absence of diffusion. The fourth condition provides the necessary requirement for diffusion-driven instability.

For the interior equilibrium to be biologically meaningful, its positivity requires:

$$\begin{cases} ES < 1, \\ \dfrac{A(ES+E-1)(BES-B+E)}{(ES-1)^3} > 0. \end{cases}$$

We now show that the fourth condition in (13) fails for any parameter values consistent with the above. The $(2,2)$ entry of the Jacobian evaluated at the positive equilibrium is

$$a_{22} = \frac{x}{Sx+1} - E.$$

At the positive equilibrium, the y-nullcline condition gives

$$\frac{dy}{dt} = \frac{xy}{Sx+1} - Ey = y\left(\frac{x}{Sx+1} - E\right) = 0,$$

and since $y \neq 0$ at the interior equilibrium, it follows immediately that

$$a_{22} = 0.$$

Consequently, the second condition in (13) reduces to $a_{11} < 0$, and upon substituting $a_{22} = 0$, the fourth condition becomes

$$D\,a_{11} > 2\sqrt{D\ \det J_{E_4}}, \text{ or equivalently, } D\,a_{11} - 2\sqrt{D\ \det J_{E_4}} > 0.$$

However, since $D > 0$ and $a_{11} < 0$, the left-hand side is strictly negative for all admissible parameter values. Therefore, the fourth condition in (13) is never satisfied, and Turing instability cannot occur in this system.

Appendix C. Conditions for stability analysis of E_1 and E_2.

- The total extinction equilibrium is given by $E_1 = (0, 0)$. At this point, the Jacobian matrix is

  $$J_{E_1} = \begin{pmatrix} -AB & 0 \\ 0 & -E \end{pmatrix},$$

  which has two negative eigenvalues. Hence, the equilibrium E_1 is always locally stable.
- The predator-free equilibrium is given by $E_2 = (1, 0)$, which corresponds to the extinction of predators while the prey persists at its carrying capacity. At this point, the Jacobian matrix takes the form

  $$J_{E_2} = \begin{pmatrix} -A(1-B) & -\frac{1}{S+1} \\ 0 & -E + \frac{1}{S+1} \end{pmatrix}.$$

The conditions for local stability are

$$\begin{cases} -A(1-B) - E + \frac{1}{S+1} < 0, \\ -A(1-B)\left(-E + \frac{1}{S+1}\right) > 0. \end{cases}$$

Since $A > 0$ and $B < 1$, we have $-A(1-B) < 0$. Thus, stability requires

$$-E + \frac{1}{S+1} < 0 \quad \Longrightarrow \quad E > \frac{1}{S+1}.$$

From a biological standpoint, E represents the predator's per capita linear mortality rate, while S denotes the prey saturation constant. Consequently, predator extinction occurs once the mortality rate surpasses the critical value $1/(S+1)$. This outcome is biologically reasonable: when the saturation constant S is small, predators utilize prey more efficiently and their population grows more quickly, so a comparatively larger death rate is needed to drive the predator population to extinction.

Appendix D. Derivation of the network dispersion relation

Consider the network system (18). Let (x^*, y^*) denote a homogeneous equilibrium satisfying $f(x^*, y^*) = 0$ and $g(x^*, y^*) = 0$. Introduce small perturbations

$$x_i = x^* + \varepsilon\,\xi_i(t), \qquad y_i = y^* + \varepsilon\,\eta_i(t), \qquad 0 < \varepsilon \ll 1.$$

Linearizing around (x^*, y^*) gives

$$\frac{d}{dt}\begin{pmatrix}\xi_i \\ \eta_i\end{pmatrix} = J\begin{pmatrix}\xi_i \\ \eta_i\end{pmatrix} + \begin{pmatrix}D_x & 0 \\ 0 & D_y\end{pmatrix}\sum_{j=1}^{N} L_{ij}\begin{pmatrix}\xi_j \\ \eta_j\end{pmatrix}, \tag{D.1}$$

where

$$J = \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix}_{(x^*, y^*)}$$

is the Jacobian evaluated at the equilibrium. Since L is diagonalizable, let $L\phi^{(k)} = \Lambda_k \phi^{(k)}$, $k = 1, \dots, N$. Expanding the perturbations in the Laplacian eigenbasis,

$$\xi_i(t) = a_k(t)\,\phi_i^{(k)}, \qquad \eta_i(t) = b_k(t)\,\phi_i^{(k)},$$

decouples (19) into independent modes. For each eigenmode Λ_k,

$$\frac{d}{dt}\begin{pmatrix}a_k \\ b_k\end{pmatrix} = \left(J - \Lambda_k\begin{pmatrix}D_x & 0 \\ 0 & D_y\end{pmatrix}\right)\begin{pmatrix}a_k \\ b_k\end{pmatrix}. \tag{D.2}$$

The characteristic equation for the growth rate λ of mode Λ_k is

$$\det\left(J - \Lambda_k\begin{pmatrix}D_x & 0 \\ 0 & D_y\end{pmatrix} - \lambda I\right) = 0. \tag{D.3}$$

Expanding the determinant (D.3) in yields Eq. (19) in the main text.

Data availability

No data was used for the research described in the article.

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